



DESIGN AND SIMULATION OF A SIMPLIFIED PCI EXPRESS GEN 6 DIGITAL MODEL

Hossam Fawzy Abdel-moniem

Arwa Hany Saeed

Youssif Mahmoud Shapana

Ahmed Mohamed Sanad

Ahmed Alaa Gafaar

Under Supervision of

DR. Omar Mahmoud Sabry

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Abbreviations

PCIe	Peripheral Component Interconnect Express
NRZ	Non-Return-to-Zero
PAM4	Pulse Amplitude Modulation
FRS	Fast Replay System
TLP	Transaction Layer Packet
DLLP	Data Link Layer Packet
PHY	Physical Layer
FLIT	Flow Control Unit
FEC	Forward Error Correction
LCRC	Link Cyclic redundancy check
ECRC	End-to-End CRC
CRM	Cyclic Redundancy Mechanism
BER	Bit Error Rate
SER	Symbol Error Rate
SNR	Signal-to-Noise-Ratio
ACK	Acknowledgment
NAK	Negative Acknowledgment
FC	Flow Control

Abstract

The continuous expansion of data-driven applications and high-performance computing systems has increased the demand for faster, more reliable communication between internal computer components. **The Peripheral Component Interconnect Express (PCIe)** standard has become a fundamental technology in modern systems, providing a high-speed pathway that connects processors with devices such as graphics cards, storage units, and specialized accelerators. This study presents a detailed overview of the evolution of PCIe, from its early generations to the latest release, PCIe 6. The new generation introduces several major improvements that significantly enhance system performance. These include much higher data transfer capabilities, more efficient methods for organizing data, improved signal handling, and stronger mechanisms to ensure data accuracy during transmission. Together, these enhancements allow information to move more quickly, efficiently, and reliably across the system. By analyzing these advancements, this work demonstrates how PCIe 6 meets the increasing requirements of modern technologies, including artificial intelligence systems, high-speed data storage, and large-scale data-center infrastructure. In conclusion, PCIe 6 represents a significant step forward in interconnect technology, supporting the next wave of innovation in computing performance, system scalability, and overall efficiency.

Chapter One: Introduction

This chapter provides a brief study of the Peripheral Component Interconnect Express (*PCIe*) standard, presenting its background, fundamental concepts, and layered architecture. The chapter explains the limitations of traditional bus-based interconnects and describes how PCIe evolved to address increasing bandwidth and performance demands. It also presents the progression of PCIe generations from Gen 1 through Gen 5, leading to the motivation behind PCIe Gen 6. Key architectural enhancements and features introduced in PCIe Gen 6 are highlighted, particularly those enabling higher data rates and improved reliability. Finally, the chapter presents the problem statement, project objectives, and scope, establishing the foundation for the simulation-based study of PCIe Gen 6 in the subsequent chapters.

1.1. Background

Traditional bus-based interconnects like PCI and AGP were unable to keep pace with the rising demands of modern computing. Their limitations in bandwidth, scalability, and efficiency highlighted the need for a more advanced communication interface. PCI Express (PCIe) was introduced to overcome these constraints by providing a high-speed, low-latency, and scalable architecture capable of supporting the performance requirements of contemporary systems.

1.2 Overview of PCIe Express

Peripheral Component Interconnect Express (PCIe) is a high-speed serial computer expansion interface standard used to connect major hardware components inside a computer system. It serves as the primary communication link between the central processing unit (CPU), the chipset, and high-performance peripheral devices such as graphics cards (GPUs), solid-state drives (NVMe SSDs), network interface cards, and other expansion devices.

In simple terms, PCIe can be described as the main internal data highway of modern computers, enabling fast and efficient data transfer between the processor and connected devices. Earlier interconnect standards such as PCI and PCI-X relied on parallel communication, where multiple data bits were transmitted simultaneously. As operating frequencies increased, these parallel buses suffered issues such as: Signal skew, Electromagnetic interference, Limited scalability, and Reduced reliability at high speeds. To overcome these limitations, PCIe was introduced as a serial, point-to-point communication standard, allowing higher data rates, better signal integrity, and improved system scalability.

PCIe uses a **point-to-point topology**, meaning that each device has a **direct connection** to the root complex (usually the CPU or chipset), rather than sharing a common bus with other devices. This design eliminates contention between devices and significantly improves overall system performance.

1.2.1 PCIe TOPOLOGY

PCIe Topology will help to understand how different peripheral devices will communicate with the CPU through the PCIe link. The CPU will access the root complex to configure the IP, and CPU will also configure the different endpoints with the help of the root complex. The root complex can directly access the memory. Thus, the PCIe endpoints will communicate the CPU and the memory with the help of root complex as shown in **Figure1**. We can attach many PCIe endpoints with the root complex with the help of PCIe Switches. These different components will communicate between the CPU or Memory and peripheral devices through the PCI express expansion bus. These PCIe endpoints can read and write data from memory. [1]

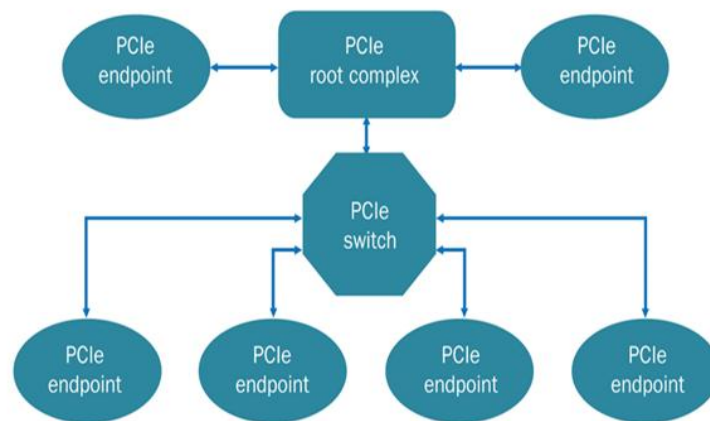


Figure 1:The PCIe Topology [1]

1.2.2 PCIe ARCHITECTURE

The PCIe bus is a serial bus expansion which works on three different layers out of the seven layers of the OSI model as shown in **Figure2**. These three layers are physical layer, data link layer, and transaction layer. The data is passed through these layers. The mechanism of PCIe standard is plug and play based PCI. The Physical layer provides the point-to-point serial connection between the peripheral devices and processor unit of PCI. The read and write requests of transportation occurred in the transaction layer. The mechanism of the PCIe depends upon this layer. The packet-based transaction and split transaction protocol occurred in the transaction layer. The link layer is used to link the sequence numbers and CRC of the packets for the transaction layer. This link provides a highly reliable data transfer mechanism. The physical layer has a transmit pair and a receive pair. The combination of transmit pair and the receive pair is called a lane. The bandwidth of the PCI Express Lane is about 250MB/s in each direction. Nowadays, the PCI board data rate can be increased by folding it twice or fourth times. This bandwidth is providing for the same devices. [1]

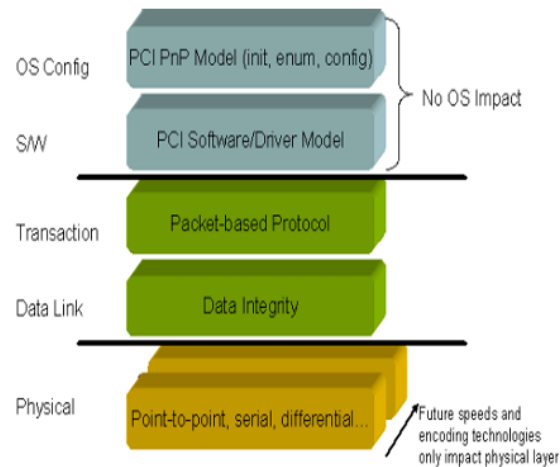


Figure 2:The PCI Express Architecture [1]

1.3 Evolution of PCIe Generations

PCIe has evolved through multiple generations, each introducing advancements in speed, bandwidth, and features (**Table1**): [2]

Revision**	Max Data Rate	Encoding	Signaling
PCIe 8.0 (TBD)	Pathfinding	TBD	TBD
PCIe 7.0 (2025)	128.0 GT/s	1b/1b (Flit Mode*)	PAM4
PCIe 6.0 (2022)	64.0 GT/s	1b/1b (Flit Mode*)	PAM4
PCIe 5.0 (2019)	32.0 GT/s	128b/130b	NRZ
PCIe 4.0 (2017)	16.0 GT/s	128b/130b	NRZ
PCIe 3.0 (2010)	8.0 GT/s	128b/130b	NRZ
PCIe 2.0 (2007)	5.0 GT/s	8b/10b	NRZ
PCIe 1.0 (2003)	2.5 GT/s	8b/10b	NRZ

Table 1:Evolution of PCIe improvements [2]

Gen1: PCI Express Generation 1 was the first version of the PCIe standard, introduced in the early 2000s. It operates at a data rate of 2.5 GT/s (gigatransfers per second) per lane, using 8b/10b encoding, which adds overhead for signal integrity. This gives an effective bandwidth of about 250 MB/s per lane (e.g., ~4 GB/s on an x16 link). Gen1 laid the foundation for PCIe’s scalable, serial-link design, replacing older parallel buses like PCI.

Gen2: PCI Express Generation 2 doubled the data rate of Gen1 to 5.0 GT/s per lane, while still using 8b/10b encoding. This results in an effective bandwidth of roughly 500 MB/s per lane (about 8 GB/s on an x16 link). Gen2 improved overall performance and supported higher-speed peripherals while maintaining backward compatibility with Gen1 devices and slots.

Gen3: Introduced in 2010, Gen3 operates at a maximum data rate of 8 GT/s (gigatransfers per second), achieving an effective bandwidth of approximately 1 GB/s per lane. It employs a 128b/130b encoding scheme, which significantly reduces overhead compared to earlier PCI standards. [3]

Gen4: Launched in 2017, Gen4 doubled the data rate to 16 GT/s, providing a bandwidth of about 2 GB/s per lane. This generation also introduced enhancements in signal integrity and power management, enabling more efficient communication. [3]

Gen5: Released in 2019, Gen5 further increased the data rate to 32 GT/s, effectively doubling the bandwidth to 4 GB/s per lane. This generation introduced new features such as improved error correction and enhanced power efficiency.[3]

Gen7: PCI Express Generation 7 (PCIe 7.0) is the upcoming evolution of the PCIe high-speed expansion bus, targeting a data rate of up to 128 GT/s (gigatransfers per second) per lane — double that of PCIe 6.0. When used in a typical x16 configuration, this enables up to about 512 GB/s of bidirectional raw bandwidth before accounting for encoding overhead. Gen7 continues to use PAM4 (Pulse Amplitude Modulation with 4 levels) signaling and flit-based encoding (introduced in Gen6) to maximize transmission efficiency and power performance, while maintaining backward compatibility with previous PCIe versions. It's aimed at high-performance applications such as AI/ML, cloud and data-center computing, and other data-intensive systems.[2]

1.4 Motivation for PCIe Gen 6

Before 2015, PCIe provided more bandwidth than typical applications required, but rising global data traffic made previous generations insufficient. With data centers moving to 100G, 400G, and 800G Ethernet, PCIe became a performance bottleneck. PCIe Gen 6.1 solves this by offering up to 128 GB/s on a $\times 16$ link, supporting modern data-center networks and full-duplex operation. The growing demands of AI/ML workloads also benefit from faster data movement, improving training speed, reducing latency, and enhancing large-scale system efficiency. **(Figure3)**

Ultimately, PCIe Gen 6 is essential for modern networking, AI acceleration, and high-performance computing infrastructure.

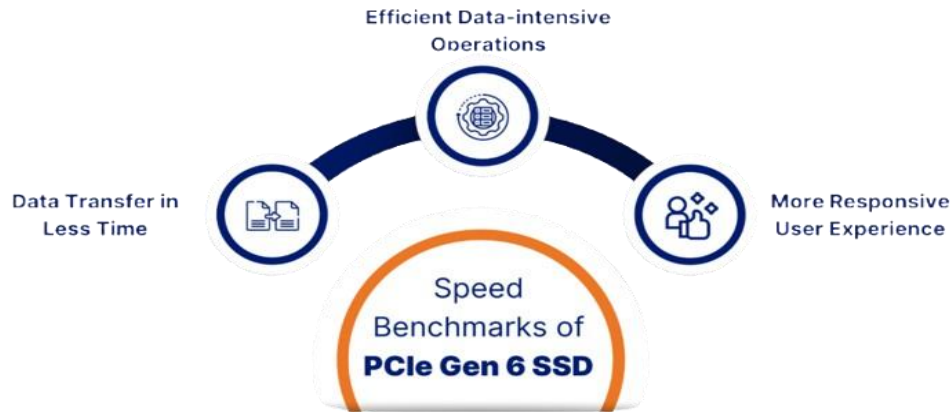


Figure 3: Speed Benchmarks and Practical Implications. [25]

1.5 Key Enhancements and Architectural Changes

PCIe 6.0 doubles the data rate of PCIe 5.0 to **64 GT/s per lane**, significantly boosting bandwidth and performance. It switches to **PAM-4 signaling** to encode more bits per symbol and uses **Forward Error Correction (FEC)** to improve data reliability at high speeds. The standard maintains **backward compatibility** with earlier PCIe versions, easing integration into existing systems. PCIe 6.0 also includes power efficiency improvements and enhanced link power management, which help reduce energy use during idle or low activity states.

These enhancements make PCIe 6.0 suitable for future high-bandwidth systems.



Figure 4:PCIe Gen 6.0 specification features. [25]

1.6 Problem Statement

PCIe 6.0 achieves a significant bandwidth increase by operating at 64 GT/s per lane using PAM4 signaling; however, this advancement introduces several critical challenges. PAM-4 reduces signal voltage margins, making links more vulnerable to noise, crosstalk, jitter, and channel loss, which degrades signal integrity compared to earlier NRZ-based generations. To address the resulting higher bit error rates, PCIe 6.0 mandates Forward Error Correction (FEC), which improves reliability but adds latency and protocol overhead, creating performance trade-offs for latency-sensitive applications. Additionally, the Physical Layer becomes substantially more complex due to advanced equalization, clock recovery, and error-handling mechanisms, leading to increased design difficulty, power consumption, and thermal concerns. These issues collectively make the implementation, verification, and optimization of PCIe 6.0 more challenging despite its substantial throughput advantages.

1.7 Project Objectives

The primary objective of this project is to study, model, and simulate the PCI Express (PCIe) Generation 6 protocol in order to understand its layered architecture, data transfer mechanisms, and performance improvements over earlier PCIe versions. The project aims to develop a simplified, simulation-based hardware model that demonstrates how PCIe Gen 6 manages data transactions between a host and peripheral devices, including key operations such as link training, packet generation and transmission, and error detection and correction. The scope of the work is focused on representing the logical system architecture and data flow across the three main PCIe layers—Transaction Layer, Data Link Layer, and a behavioral digital model of the Physical Layer—while intentionally avoiding unnecessary complexity related to full physical hardware implementation. Using hardware description and simulation tools such as Verilog, ModelSim, or equivalent environments, the project bridges theoretical protocol concepts with practical implementation, providing an educational and analytical insight into PCIe Gen 6 functionality and behavior at the system level.

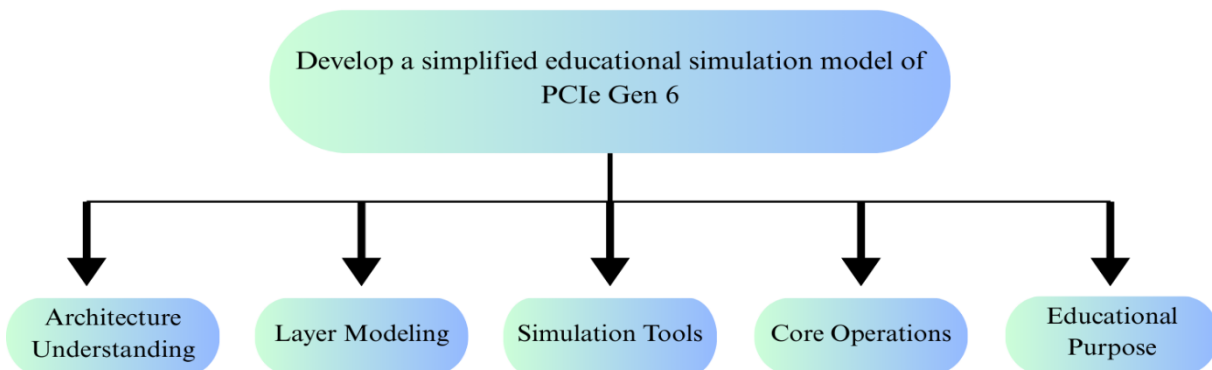


Figure 5:PCIe Gen 6.0 Objectives.

Chapter Two: Previous studies

This chapter presents a comprehensive study of previously related work to the PCI Express (PCIe), aiming to establish the theoretical and technical background for the proposed work. The reviewed literature covers key aspects of PCIe, including architecture, generational development, physical-layer signalling techniques, packet framing mechanisms, error control strategies, and existing implementations. Special attention is given to recent advancements such as FLIT-based packet framing, PAM4 modulation, the transition from NRZ signalling, and the integration of FEC and CRC in high-speed links. The chapter also discusses verification challenges associated with PCIe Gen6 and concludes by identifying key research gaps that motivate the objectives and contributions of this project.

2.1 Studies About PCIe Architecture

Understanding the PCI Express (PCIe) architecture is essential for analyzing and designing high-speed interconnect systems. A clear architectural overview enables deeper comprehension of PCIe protocol behavior, data flow, and error-handling mechanisms used in modern computing platforms. Consequently, numerous studies have been published addressing PCIe architecture from different perspectives, including performance, reliability, and fault tolerance.

One of the most significant and relevant studies is “**Fault-Resilient PCIe Bus with Real-time Error Detection and Correction**”, published in May 2021 by Mostafa Darvishi, Ph.D., Member IEEE, and an independent high-performance computing researcher.[4]

This work is considered a reliable and authoritative reference due to its detailed architectural analysis and its practical implementation on FPGA-based PCIe systems and its crucial role in past and current research.

The paper presents a comprehensive description of the **three-layer PCIe architecture**, which consists of the Transaction Layer (TL), Data Link Layer (DLL), and Physical Layer (PL) as shown in **Figure6**. The Transaction Layer is responsible for packet formation, managing end-to-end data transfers, and detecting errors such as unsupported requests, completion timeouts, and corrupted Transaction Layer Packets (TLPs). The Data Link Layer ensures reliable packet delivery by performing Link CRC (LCRC) checks, sequence number verification, and retransmission in case of detected errors. The Physical Layer manages link training, lane initialization, and low-level electrical signaling, ensuring stable high-speed communication between devices.

In addition to layer functionality, the study explains the PCIe lane-based architecture, where PCIe links are implemented using scalable lane configurations such as x1, x4, x8, x16, and x32. Increasing the number of lanes allows higher throughput by transmitting data in parallel across multiple differential pairs. The paper also characterizes PCIe as a point-to-point, dual-simplex, high-speed serial interconnect that relies on differential signaling to improve noise immunity and signal integrity.

Furthermore, the paper surveys several system-level PCIe architectural studies, including DMA-based PCIe architectures, bridging mechanisms, these studies highlight how PCIe can be adapted for different applications such as high-performance computing, data acquisition systems, and security analysis as shown in **Figure6**.

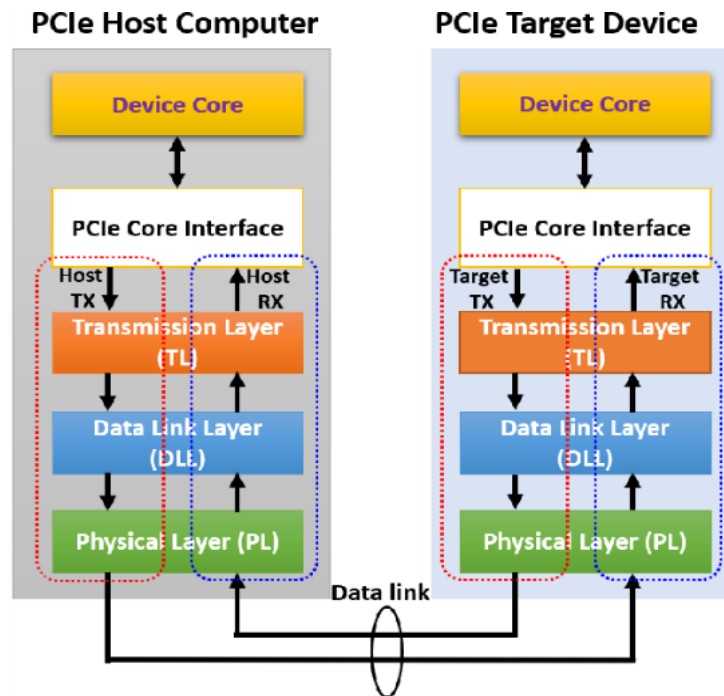


Figure 6:PCIe architecture overview [4]

Beyond basic architectural concepts, the authors emphasize the importance of error classification and reliability mechanisms within PCIe systems. Errors are analyzed at each architectural layer, and the paper categorizes them into correctable, non-correctable, non-fatal, and non-correctable fatal errors, illustrating how PCIe maintains system stability even under fault conditions.

Overall, this study provides a solid architectural foundation for understanding PCIe operation and reliability. Its detailed discussion of protocol layers in **Figure6**, lane scalability, signaling characteristics, and error-handling mechanisms makes it a fundamental reference for research and development in high-speed PCIe-based systems.

Moreover, this architecture was used in many articles and books afterwards by another researchers and publishers due to its simplicity and clarity so it was prementioned in another authorized study “**Technical Analysis of PCIe to PCIe 6: A Next-Generation Interface Evolution**” that was published in August 2023 by Santhosh Nagaraj Nag, which provides a systematic comparison of PCIe generations from Gen1 through Gen6.[5]

Each PCIe lane consists of two differential signaling pairs: one for transmission (TX) and one for reception (RX). Lanes are unidirectional and can be combined to form wider links, such as $\times 1$, $\times 4$, $\times 8$, or $\times 16$, depending on bandwidth requirements. This lane-based structure allows PCIe to scale performance while maintaining signal integrity and reliability across generations.

Physically, PCIe devices connect through standardized slots that support different lane configurations and form factors. The number of lanes associated with a slot directly determines the available bandwidth, with larger configurations offering higher throughput. PCIe also supports backward and forward compatibility, enabling devices and hosts of different generations to interoperate at the highest mutually supported link speed and width.

Logically, PCIe systems follow a hierarchical organization centered around a host device, typically the CPU or chipset as clarified in **Figure 7**. The host contains one or more PCIe Root Complexes responsible for device initialization, configuration, and transaction management. Peripheral devices communicate with the host using packet-based protocols, while PCIe switches may be used in larger systems to expand connectivity by enabling multiple endpoints to share a single root complex.

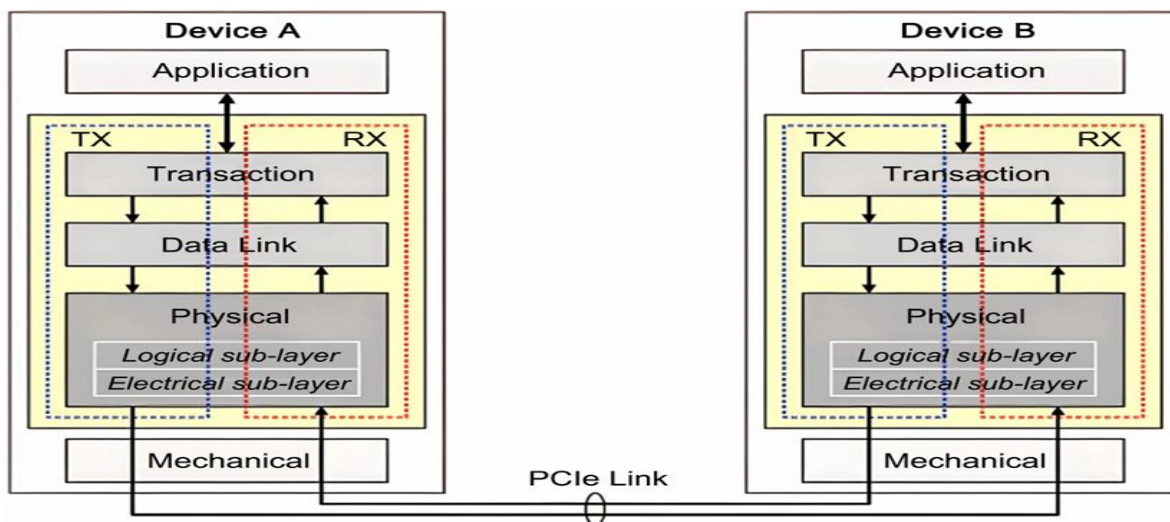


Figure 7:PCIe architecture overview [5]

2.2 Studies About PCIe Generational Development

The evolution of Peripheral Component Interconnect Express (PCIe) was driven by the continuous increase in processor performance and the growing demand for higher data transfer rates between system components. Before PCIe, conventional PCI architectures relied on a shared parallel bus **Figure 8**, which became a significant performance bottleneck as multiple devices competed for the same bandwidth. This limitation motivated the transition toward a high-speed, point-to-point serial interconnect, leading to the introduction of PCIe in the early 2000s.

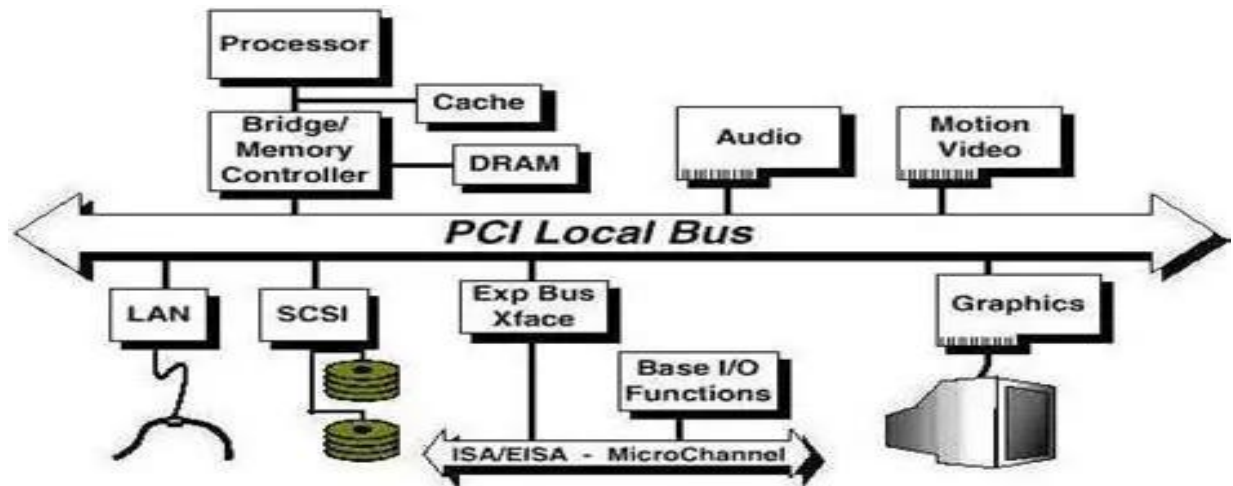


Figure 8:old PCI bus [6]

Unlike legacy PCI, PCIe introduced a scalable lane-based architecture and dedicated point-to-point links, allowing each connected device to access its own bandwidth independently. However, early versions of PCIe were not designed to support the extremely high data rates required by modern processors, accelerators, and storage devices. As a result, PCIe has undergone continuous generational development, with each new generation addressing performance limitations, signal integrity challenges, and protocol efficiency issues observed in previous versions.

Numerous studies have analysed the generational evolution of PCIe in detail. One of the most comprehensive references is the paper “**Technical Analysis of PCIe to PCIe 6: A Next-Generation Interface Evolution**”, published in August 2023 by Santhosh Nagaraj Nag, which provides a systematic comparison of PCIe generations from Gen1 through Gen6, focusing on data rate scaling, encoding techniques, and protocol enhancements.[5]

According to PCI-SIG. (2002) “**PCI Express Base Specification, Revision 1.0**” in April 29, 2002, **PCIe Gen1** established the foundation of the PCIe architecture by supporting a data rate of **2.5 GT/s per lane**, with scalable lane configurations ranging from x1 to x16. This generation introduced the fundamental concepts of point-to-point communication, and backward compatibility. (PCIe Gen1) introduced a scalable, packet-based interconnect architecture that replaced legacy shared-bus designs. It defined structured transaction types and strict ordering rules, enabling reliable and predictable data transfer across point-to-point links within the PCIe fabric. [6]

These features established the foundation for consistent communication and extensibility in subsequent PCIe generations. Power efficiency and system flexibility were central to PCIe Gen1, with support for multiple device power states, software-controlled power transitions, and platform-level power budgeting. The standard also incorporated Quality of Service (QoS) mechanisms to enable differentiated traffic handling, as well as hot-plug and hot-swap support and advanced peer-to-peer communication across multiple hierarchies, allowing scalable and dynamic system expansion.

According to “M. Tj, “**Difference Between PCI Express Gen 1, Gen 2, Gen 3, Gen 4, Gen 5, & Gen 6,**” Desk Decode” in Jul. 6, 2017. **PCIe Gen2** built upon the architectural foundations established in Gen1 while significantly increasing link performance. The primary enhancement introduced in Gen2 was the doubling of the raw data rate from 2.5 GT/s to 5.0 GT/s per lane, resulting in substantially higher bandwidth without altering the fundamental packet-based protocol. PCIe Gen2 maintained backward compatibility with Gen1 devices, allowing systems to negotiate link speed and width dynamically based on the capabilities of connected components.[7]

In addition to higher throughput, **PCIe Gen2** introduced improvements in signal integrity and power management to support increased data rates. Enhanced link equalization, refined clocking mechanisms, and optimized electrical specifications enabled reliable operation at higher speeds. The generation also preserved and extended Quality of Service (QoS), hot-plug support, and advanced error handling features, ensuring scalability and robustness in performance-critical and high-density computing systems. As shown in **Figure9**.

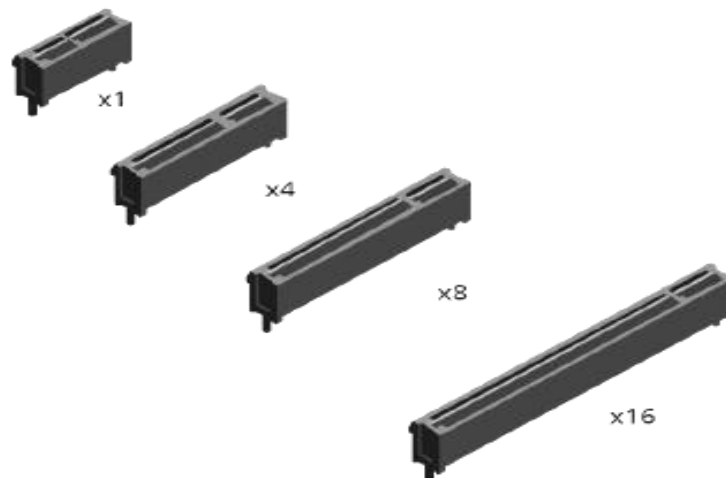


Figure 9:PCIe X1, X4, X8, X16 from different generations [7]

PCIe Gen3 represented a major architectural improvement as shown by **PCISIG in ‘PCI Express Base Specification Revision 3.0’** in November 2010, PCI Express Generation 3 (PCIe Gen3) enhanced the data rate to **8 GT/s per lane** PCIe and architecture by improving efficiency, reliability, and service differentiation while preserving backward compatibility. It continued to support structured transactions and ordering rules and expanded Quality of Service (QoS) capabilities through end-to-end packet tagging, configurable arbitration, and improved handling of time-sensitive data flows, reducing head-of-line blocking in complex systems.[8]

PCIe Gen3 also strengthened power management, data integrity, and error handling mechanisms. Devices can support advanced power-state control, wake-up signalling, and platform-level power budgeting, alongside unified hot-plug and hot-swap support. Enhanced link-level and end-to-end error detection, advanced error reporting, and improved testability further increased the reliability and scalability of PCIe Gen3-based systems.

Figure10 illustrates the PCI Express (PCIe) packet structure and the distribution of responsibilities across the protocol layers. The Transaction Layer generates the packet header and optional data payload, with optional End-to-End CRC (ECRC) providing additional error protection. The Data Link Layer adds a sequence number and a Link CRC (LCRC) to support reliable transmission through error detection and replay mechanisms. Finally, the Physical Layer applies framing to delimit packets and enable correct transmission over the serial link, ensuring proper alignment and data integrity.

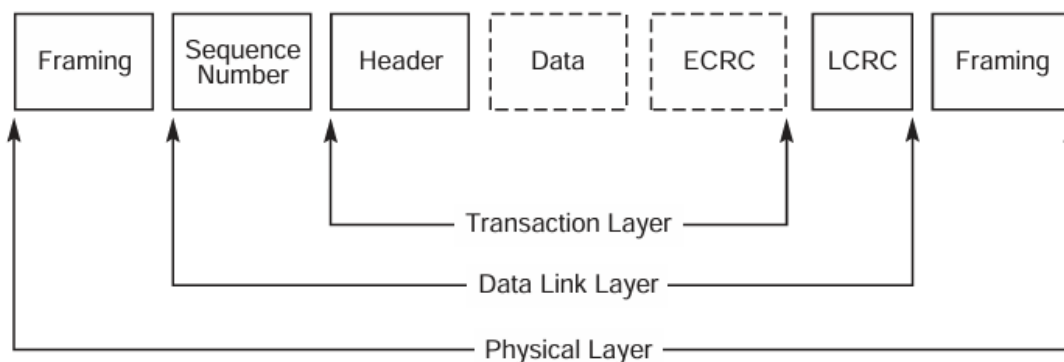


Figure 10:Packet Flow Through the Layers in GEN3 [8]

Subsequent generations focused primarily on bandwidth scaling to meet the requirements of high-performance computing and data center applications. Regarding **‘PCI Express Base Specification Revision 4.0’** which is the official reference for the **PCIe Gen 4** that was released in February 2014, PCIeV4 is designed to provide data rate of 16 GT/s per lane low-latency, high-bandwidth communication while maximizing link efficiency and application payload throughput. This is achieved through higher bandwidth per lane, scalable performance using aggregated lanes can be noticed in **Figure11**, and advanced power management capabilities, including power state control, wake-up signalling, and efficient power budgeting.[9]

The architecture also supports differentiated Quality of Service (QoS) to improve data flow efficiency and mitigate head-of-line blocking. In addition, PCIe Gen 4 introduces enhanced features such as M-PCIe support over MIPI M-PHY, native Hot-Plug functionality, improved data integrity and error handling, process technology independence, and simplified electrical compliance testing. These enhancements make PCIe Gen 4 suitable for high-performance, scalable, and reliable system designs.[10]

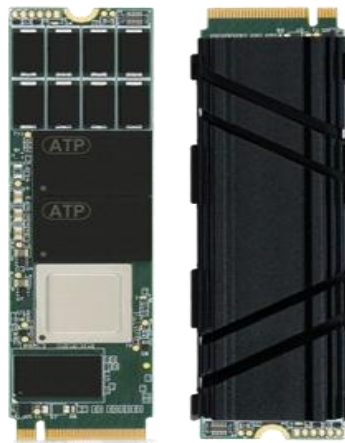


Figure 11:PCIe generation 4 by NVMe [9]

PCIe Gen5 further doubled the data rate to 32 GT/s per lane, as defined in the **PCI Express Base Specification Revision 5.0** released in May 2019, delivering substantial bandwidth improvements while maintaining backward compatibility with earlier generations. In addition to higher throughput, PCIe Gen5 extends power management capabilities by supporting system wake-up requests from low-power states and enabling controlled device power-up sequencing in accordance with platform power-budgeting policies.[11]

Moreover, PCIe Gen5 enhances Quality of Service (QoS) through differentiated services, including dedicated link resources per data flow, configurable arbitration policies, packet-level QoS tagging, and end-to-end isochronous transfers, thereby reducing head-of-line blocking and improving fabric efficiency as shown in **Figure12**. System reliability and flexibility are further strengthened through comprehensive Hot-Plug support, advanced error handling, improved data integrity, process technology independence, and simplified electrical compliance testing, ensuring robust integration within next-generation high-performance computing platforms.[12]



Figure 12:PCIe generation 5 from NVMe [11]

The most significant architectural shift occurred with **PCIe Gen6** which can be seen from ‘**PCIe 6.0 Specification: The Interconnect for I/O Needs of the Future,**’ which introduced fundamental changes beyond simple bandwidth doubling. Although Gen6 maintains a raw data rate equivalent to **64 GT/s per lane**, it abandons traditional NRZ signalling in Favor of Pulse Amplitude Modulation with four levels (PAM4). PAM4 enables the transmission of two bits per symbol, effectively increasing data throughput without doubling the clock frequency. However, this modulation scheme inherently increases the bit error rate, necessitating additional reliability mechanisms.[13]

PCIe Gen 6 introduces major architectural enhancements to maintain power efficiency and low latency. It strengthens power management through advanced wake-up mechanisms and controlled power sequencing, while expanding Quality of Service (QoS) capabilities with packet-level QoS tagging, configurable arbitration policies, and support for end-to-end isochronous traffic. These features enable predictable performance and efficient resource allocation in data-intensive and latency-sensitive workloads.[14]

In addition, PCIe Gen 6 significantly improves system robustness by integrating stronger data integrity and error-handling mechanisms, including mandatory forward error correction (FEC) and enhanced CRC protection. Native Hot-Plug support, process technology independence, and improved testability ensure scalability and reliability across diverse platforms. Collectively, these advances position PCIe Gen 6 as a foundation for next-generation accelerator, memory-centric, and high-performance computing systems.

Table2 summarizes the evolution of PCI Express (PCIe) across generations, showing how performance has increased over time. Each new PCIe version roughly doubles the data rate per lane, leading to a corresponding increase in total x16 bandwidth. Early generations (PCIe 1.0 and 2.0) used 8b/10b encoding, which introduced higher overhead, while PCIe 3.0 and later adopted 128b/130b encoding to improve efficiency. PCIe Gen6 marks a major shift by introducing PAM4 signaling and FLIT-based data transfer, enabling extremely high bandwidth while maintaining reliability.

PCIe ppecification	Year	Data Rate (Gb/s) Encoding	x16 Bandwidth*
1.0	2003	2.5 (8b/10b)	32 Gb/s
2.0	2007	5.0 (8b/10b)	64 Gb/s
3.0	2010	8.0 (128b/130b)	126 Gb/s
4.0	2017	16.0 (128b/130b)	252 Gb/s
5.0	2019	32.0 (128b/130b)	504 Gb/s
6.0	2021	64.0 (PAM-4, FLIT)	1024 Gb/s

Table 2:Next-Generation Interface Evolution [5]

To address these challenges, PCIe Gen6 introduced Forward Error Correction (FEC) to detect and correct errors without requiring retransmission, thereby maintaining low latency and high efficiency. In addition, Gen6 adopted a Fixed-Length Unit (FLIT)-based architecture, replacing variable-length packets to simplify error correction, improve latency predictability, and support high-speed operation. These changes represent a shift toward a more deterministic and robust protocol design suitable for future computing systems.

Beyond raw speed improvements, several studies emphasize that PCIe evolution also focused on reducing protocol overhead and improving real-world performance. For instance, the transition from 8b/10b to 128b/130b encoding significantly increased effective bandwidth without increasing the transfer rate. Furthermore, advanced features such as automatic lane margining, introduced in later generations, enable continuous monitoring of signal quality during normal operation, which is critical as eye diagrams shrink at higher data rates. Collectively, these studies demonstrate that PCIe generational development is driven not only by higher transfer rates but also by smarter protocol design and enhanced reliability mechanisms.[14]

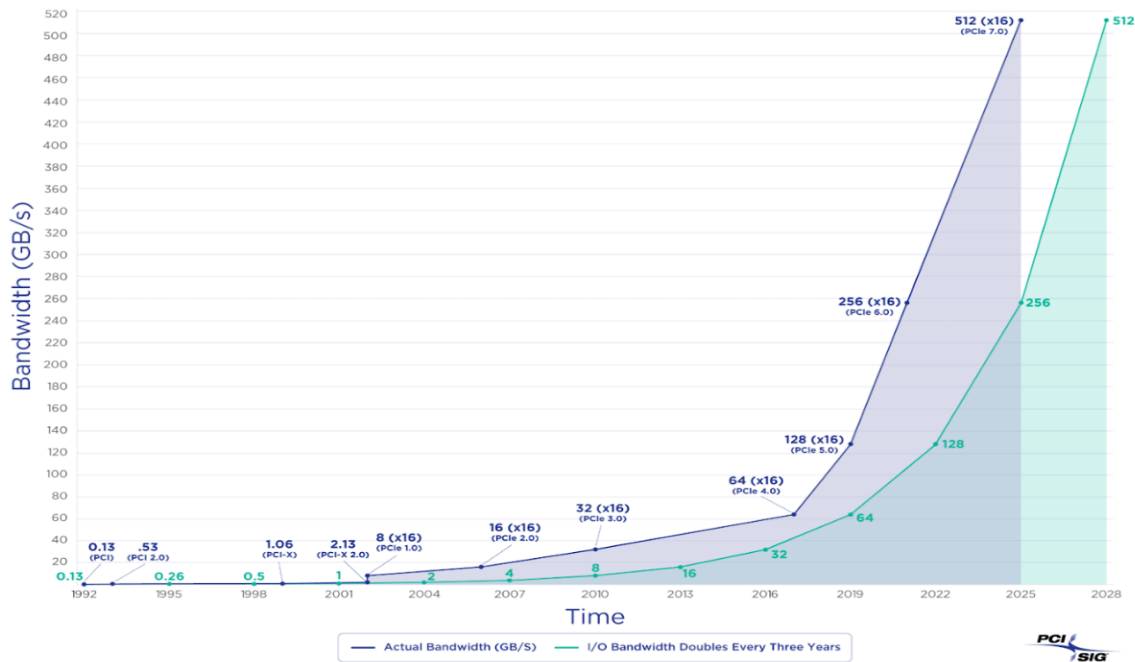


Figure 13:generational development of PCIe [13]

In addition to speed improvements across PCIe generations, several research papers focused on protocol efficiency and how overhead reduction contributed to real-world performance. For example, the transition from 8b/10b encoding in Gen1/Gen2 to 128b/130b encoding in Gen3 significantly reduced bandwidth loss and increased effective throughput even without changing the symbol rate.

Studies also analyzed the development of automatic lane margining—a feature introduced to continuously evaluate signal quality during active operation. This technique became increasingly important starting from PCIe Gen5 because of the dramatically reduced eye-diagram width. These findings highlight that PCIe evolution was not only about increasing transfer rates but also about introducing smarter and more efficient protocol mechanisms.

2.3 Studies About FLIT Mode (Fixed-Length Packets)

Several recent studies have focused on the introduction of Fixed-Length Packet (FLIT) mode, which represents one of the most significant architectural changes in PCIe Gen6. A key reference in this area is the study published in **PCIe verification at high speeds: challenges, solutions, and future trends** February 2025, which provides a detailed technical analysis of FLIT integration, verification challenges, high-speed operation considerations and FLIT layout as shown in **Figure14**. As a recent publication, this work reflects the latest developments in PCIe Gen6 protocol design and verification methodologies.[16]

With the transition to PCIe Gen6, the data transmission model shifted from variable-length Transaction Layer Packets (TLPs), used in PCIe Gen1 through Gen5, to a fixed-length FLIT-based architecture. In earlier generations, TLPs and Data Link Layer Packets (DLLPs) varied in size, providing flexibility but increasing complexity at higher data rates. At 64 GT/s with PAM4 signaling, variable-length packets become difficult to manage due to tighter timing margins and increased sensitivity to bit errors. To address these challenges, PCIe Gen6 transmits all data using fixed-size 256-byte FLITs, enabling more predictable timing, simplified framing, and efficient error correction.

Description		Lane															
		0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
16 Symbol times	TLP Bytes [0...223]	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
		16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31
		32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47
		48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63
		64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79
		80	81	82	83	84	85	86	87	88	89	90	91	92	93	94	95
		96	97	98	99	100	101	102	103	104	105	106	107	108	109	110	111
		112	113	114	115	116	117	118	119	120	121	122	123	124	125	126	127
		128	129	130	131	132	133	134	135	136	137	138	139	140	141	142	143
		144	145	146	147	148	149	150	151	152	153	154	155	156	157	158	159
		160	161	162	163	164	165	166	167	168	169	170	171	172	173	174	175
		176	177	178	179	180	181	182	183	184	185	186	187	188	189	190	191
		192	193	194	195	196	197	198	199	200	201	202	203	204	205	206	207
		208	209	210	211	212	213	214	215	216	217	218	219	220	221	222	223
	TLP, DLP	224	225	226	227	228	229	230	231	232	233	234	235	DLP 0	DLP 1	DLP 2	DLP 3
	DLP, CRC, ECC	DLP 4	DLP 5	CRC 0	CRC 1	CRC 2	CRC 3	CRC 4	CRC 5	CRC 6	CRC 7	ECC1 [0]	ECC2 [0]	ECC0 [0]	ECC1 [1]	ECC2 [1]	ECC0 [1]

Figure 14: Flit Layout in a .x16 Link [16]

The adoption of FLIT mode introduces new packet formats and header structures, increasing protocol complexity, and verification requirements. Since FLITs may encapsulate multiple TLPs or DLLPs, or alternatively split a single TLP across multiple FLITs, correct handling of FLIT boundaries is critical. Verification must ensure proper packet segmentation, reassembly, sequencing, and ordering, as errors in these processes can result in data corruption or system instability.

Furthermore, FLIT mode interacts closely with other PCIe Gen6 features, including PAM4 signaling, Forward Error Correction (FEC), and new power management states. Ensuring correct interaction between these mechanisms is a major verification challenge. For example, FEC must operate efficiently on fixed-size FLITs to detect and correct errors without introducing excessive latency, while power state transitions must not disrupt FLIT transmission or ordering. These interactions require comprehensive verification strategies to validate system reliability under various operating conditions.

The **IJCET** study emphasizes that combining FLIT mode with PAM4 signaling creates a unique verification environment that demands specialized test methodologies. The fixed-length structure simplifies certain aspects of protocol handling but introduces new requirements for validating data integrity, buffer management, and link-layer coordination. Careful verification is needed to ensure seamless interaction between the Transaction Layer and Data Link Layer when operating under FLIT-based framing.[17]

That study proves the research '**Flit-Reservation Flow Control**' posted in January 2000. indicating that FLIT mode was introduced not only to support higher data rates but also to standardize packet handling across protocol layers, ultimately simplifying controller design and improving scalability. However, this standardization comes at the cost of increased verification complexity, especially in scenarios involving FLIT multiplexing, buffer overflow prevention, and strict ordering enforcement. These challenges are expected to remain relevant in future PCIe generations, including Gen7, where FLIT-based communication is anticipated to be the default operating mode.

Flit-reservation flow control is an advanced flow-control technique in which small control flits are transmitted ahead of data flits to reserve buffer space and channel bandwidth along the communication path. By performing routing and arbitration decisions before the arrival of data, this approach enables data flits to be forwarded immediately upon arrival, eliminating unnecessary waiting time and improving buffer utilization efficiency as shown in **Figure15**

Compared to conventional flow-control mechanisms, flit-reservation significantly enhances network performance by increasing achievable throughput and reducing latency while requiring fewer buffer resources. These characteristics make it particularly well suited for high-speed and on-chip interconnection networks, where efficient resource management and low-latency communication are critical to overall system performance.

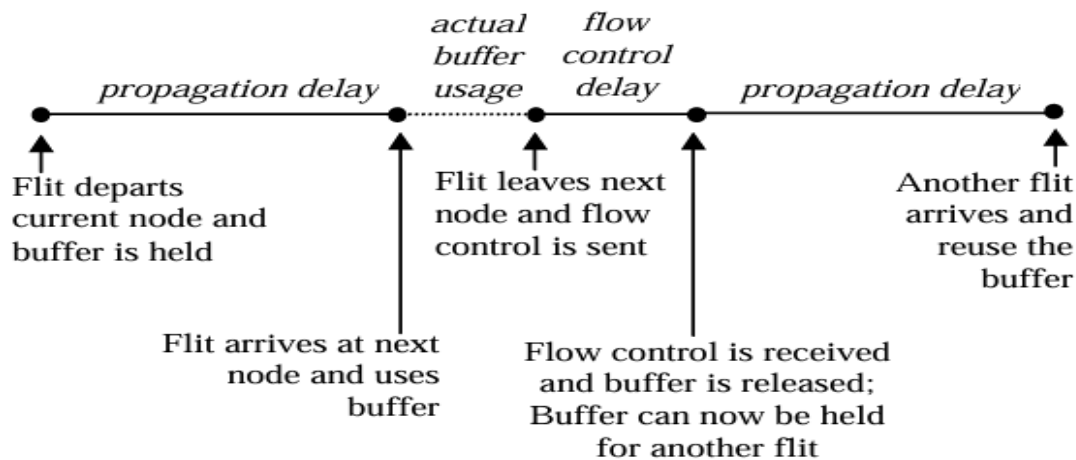


Figure 15:FLIT reservation flow [17]

2.4 Studies About PAM4 Modulation

According to the PCI-SIG educational webinar “**PCIe 6.0 Specification: The Interconnect for I/O Needs of the Future**”, presented by Debendra Das Sharma, PCIe 6.0 introduces a fundamental shift in physical-layer signaling through the adoption of PAM4 modulation to overcome the limitations of NRZ signaling at very high data rates [13].

As PCI Express continues to evolve to support the rapidly growing data-intensive demands of modern computing systems, the transition to PAM4 (Pulse Amplitude Modulation with four levels) in PCIe 6.0 represents one of the most significant changes in the physical layer signaling methodology. Previous generations of PCIe (Gen1 through Gen5) employed NRZ (Non-Return-to-Zero) signaling, also known as PAM2, which encodes one bit per symbol using two distinct voltage levels. With each generation doubling the data rate — reaching 32 GT/s in PCIe 5.0 — NRZ signaling approached its practical limits due to channel loss, increasing noise susceptibility, jitter, and the complexity required for equalization at higher symbol rates. The PCIe 6.0 specification therefore adopted PAM4 to achieve a raw data rate of 64 GT/s per lane while keeping the effective Nyquist frequency comparable to that of PCIe 5.0, thereby mitigating excessive channel loss at higher frequencies.

PAM4 signaling encodes two bits per unit interval (UI) by using four distinct voltage levels within the same period. These four levels produce three signal “eyes” within each UI, enabling higher throughput without proportionally increasing the frequency of transitions compared to NRZ signaling as can be seen in **Figure16**. However, the reduced separation between voltage levels makes PAM4 inherently more susceptible to noise and timing errors, as each eye has smaller voltage and timing margins. To address this increased vulnerability, the PCIe 6.0 specification incorporates Gray coding, which maps bit pairs such that a single symbol error results in at most a one-bit error, thereby improving robustness against noise. Additionally, precoding techniques are applied prior to transmission to reduce the impact of error bursts and enhance the effectiveness of the receiver’s processing.

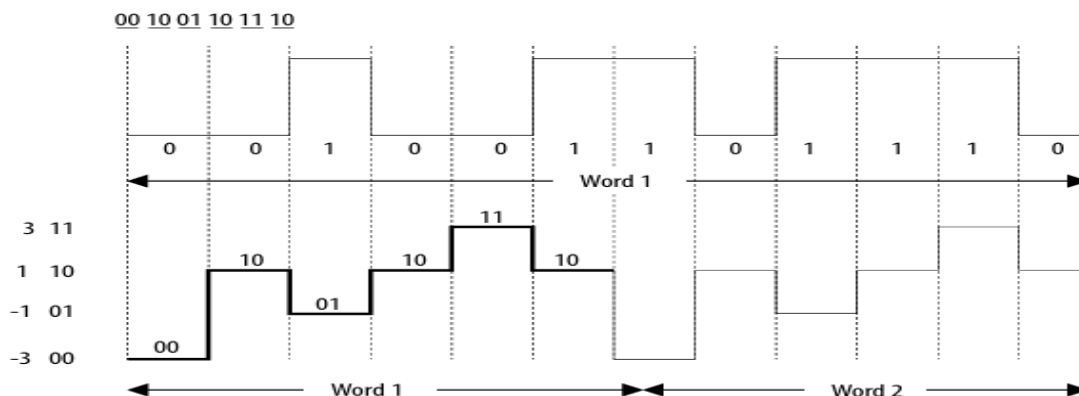


Figure 16:The new modulation technique (PAM-4) simulation [18]

Because PAM4 increases the likelihood of symbol errors compared to NRZ, PCIe 6.0 also integrates **Forward Error Correction (FEC)** in conjunction with PAM4. FEC enables the receiver to identify and correct certain bit errors without requiring retransmission, which is critical at high speeds where multiple adjacent signal levels must be reliably distinguished. These physical layer enhancements — PAM4, Gray coding, precoding, and FEC — fundamentally change the physical layer design, requiring more sensitive receiver architectures, advanced equalization techniques, and refined clock and data recovery algorithms to maintain reliable operation at 64 GT/s.

Another reference paper ‘**The Impact of PAM4 on PCIe 6.0 Electrical Testing**’ published in February 2023 discusses the electrical and verification implications of introducing four-level pulse amplitude modulation (PAM4) signaling in PCI Express Gen6. It provides an in-depth technical overview of why PAM4 was adopted, how it differs from previous NRZ signaling, and the resulting challenges in signal integrity, jitter tolerance, and electrical compliance testing.[18]

PAM4 signaling enables PCIe Gen6 to double the data rate to 64 GT/s without increasing channel bandwidth by transmitting two bits per unit interval instead of one. Unlike NRZ signaling, which produces a single eye diagram, PAM4 generates three distinct eye openings with reduced voltage margins as shown in **Figure17**. As a result, PAM4 signals are more sensitive to noise, jitter, and distortion, requiring significantly tighter electrical tolerances. The reduced eye height and width place greater demands on transmitter quality and receiver sensitivity, making accurate signal measurement and control essential for reliable operation at ultra-high speeds.

To address these challenges, PCIe Gen6 introduces enhanced equalization techniques and new electrical measurements specifically tailored for PAM4. These include stronger continuous-time linear equalization (CTLE), an increased number of decision feedback equalizer (DFE) taps, and new metrics such as signal-to-noise-and-distortion ratio (SNDR) and ratio of level mismatch (RLM). Additionally, PAM4 jitter analysis requires more complex algorithms due to multiple voltage-level transitions, as slower transitions are more susceptible to timing variation. Together, these changes ensure that PAM4-based PCIe Gen6 links can maintain reliability and low latency despite the increased signaling complexity.

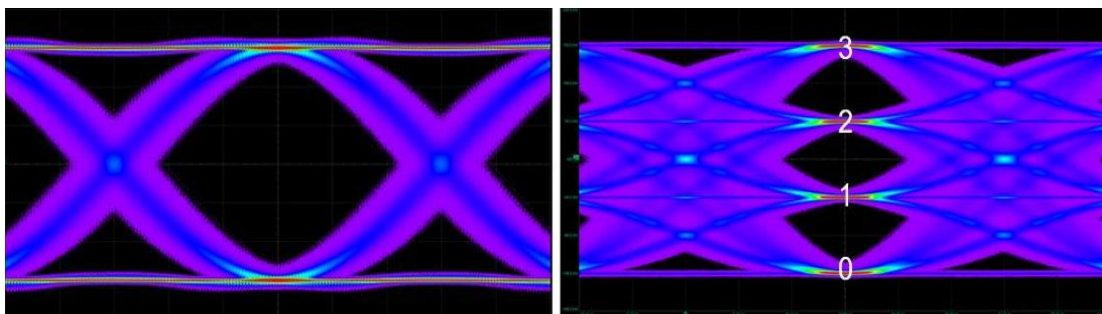


Figure 17:NRZ signal with one eye per UI, left, and PAM4 signal with three eyes, right. [19]

Overall, the transition to PAM4 in PCIe 6.0 balances the need for higher data rates with practical channel limitations and error resilience mechanisms. This evolution reflects a broader strategy within the PCI-SIG community to extend PCIe performance while preserving backward compatibility and acceptable signal integrity across real-world channel conditions.

2.5 Comparison Between NRZ and PAM4 in PCIe Generations

A major architectural shift is introduced in PCIe Gen6 through the transition from traditional NRZ (PAM2) signaling to four-level Pulse Amplitude Modulation (PAM4). According to the PCI-SIG technical analysis presented in “**PCIe 6.0 Specification: The Interconnect for I/O Needs of the Future**”, this transition was driven by the need to double the effective data rate without increasing the Nyquist frequency of the channel. While NRZ signaling encodes one bit per unit interval (UI), PAM4 encodes two bits per UI, enabling PCIe Gen6 to achieve 64 GT/s per lane while maintaining a bandwidth profile comparable to PCIe Gen5 operating at 32 GT/s.[13],

The same study highlights several critical challenges associated with PAM4 signaling. Because PAM4 utilizes four voltage levels instead of two as shown in **Figure18**, the vertical eye opening is reduced to approximately one-third of that of NRZ, significantly increasing susceptibility to noise, jitter, crosstalk, and inter-symbol interference (ISI). Consequently, the raw Bit Error Rate (BER) of PAM4 links is inherently higher than that of NRZ, making reliable error-free transmission infeasible without additional protection mechanisms.

Furthermore, the reduced voltage margins impose stringent requirements on the physical channel. Even minor channel impairments, PCB losses, or impedance discontinuities can lead to severe signal distortion. The PCI-SIG analysis emphasizes that PAM4 operation necessitates advanced equalization techniques, including Feed-Forward Equalization (FFE), Decision Feedback Equalization (DFE), and Continuous-Time Linear Equalization (CTLE). In addition, PAM4 demands highly sensitive receiver architectures capable of accurately resolving closely spaced voltage levels under high-speed operating conditions.

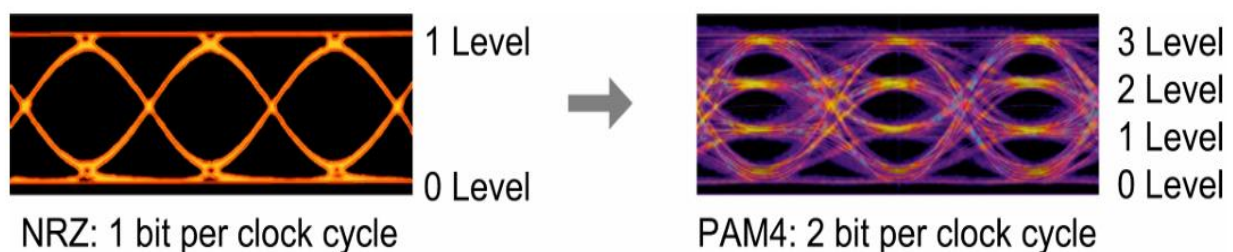


Figure 18:NRZ vs PAM4 [19]

Due to the inherently higher error probability introduced by PAM4, Forward Error Correction (FEC) is mandated as a fundamental component of PCIe Gen6. This lightweight FEC scheme is specifically optimized to correct symbol-level errors caused by PAM4 while maintaining low latency, which is essential for preserving PCIe's performance characteristics. Moreover, the adoption of FLIT-based framing in Gen6 complements FEC operation by providing fixed-size data units that align efficiently with error detection and correction processes.

Overall, the literature consistently indicates that the transition to PAM4 in PCIe Gen6 represents not merely an incremental performance enhancement but a fundamental transformation of the physical layer. This shift introduces a more complex multi-level signaling paradigm, requiring new strategies in equalization, error correction, channel design, and verification. As a result, PAM4 is regarded as one of the most technically demanding aspects of PCIe Gen6 development.

2.6 Studies About FEC + CRC in High-Speed Links

specification documents and industry-level technical analyses. A primary reference is the PCI-SIG educational webinar “**PCIe 6.0 Specification: The Interconnect for I/O Needs of the Future**”, presented by Debendra Das Sharma, which explains the architectural motivations behind introducing) and enhanced CRC mechanisms in PCIe Gen6. Complementary insight is provided by an industry analysis published by Cadence Design Systems, titled “Demystifying Forward Error Correction (FEC) in PCIe 6.0”, which focuses on the practical implications of FEC deployment in real high-speed designs.[13]

According to the PCI-SIG specification analysis, the transition to 64 GT/s per lane using PAM4 signaling significantly increases the raw Bit Error Rate (BER) due to reduced voltage margins and higher sensitivity to noise, jitter, and channel impairments. Under these conditions, traditional error handling approaches used in earlier PCIe generations—primarily CRC detection followed by packet retransmission—are no longer sufficient to guarantee reliable operation while maintaining low latency. As a result, PCIe Gen6 mandates the use of Forward Error Correction in conjunction with CRC to preserve link reliability and performance.

The PCI-SIG study emphasizes that the effectiveness of this approach is evaluated using metrics such as retry probability, Failure-In-Time (FIT) rate, and overall bandwidth efficiency. Rather than relying on strong but latency-intensive error correction, PCIe Gen6 adopts a lightweight, low-latency FEC scheme that corrects common symbol errors introduced by PAM4 while allowing rare retransmissions to handle uncorrectable errors. This design choice ensures that latency-sensitive applications are not negatively impacted by constant error correction overhead.

From **Figure19** an industry perspective, the Cadence analysis further clarifies why FEC is indispensable in PCIe Gen6 implementations. It highlights that PAM4 signaling inherently produces a higher BER compared to NRZ, making error correction mandatory for practical system deployment. The Cadence study explains that without FEC, achieving acceptable error rates at 64 GT/s would require prohibitively strict channel requirements and significantly increased signal integrity complexity. By integrating FEC, PCIe Gen6 enables reliable operation over realistic PCB channels while maintaining high throughput and manageable design costs.

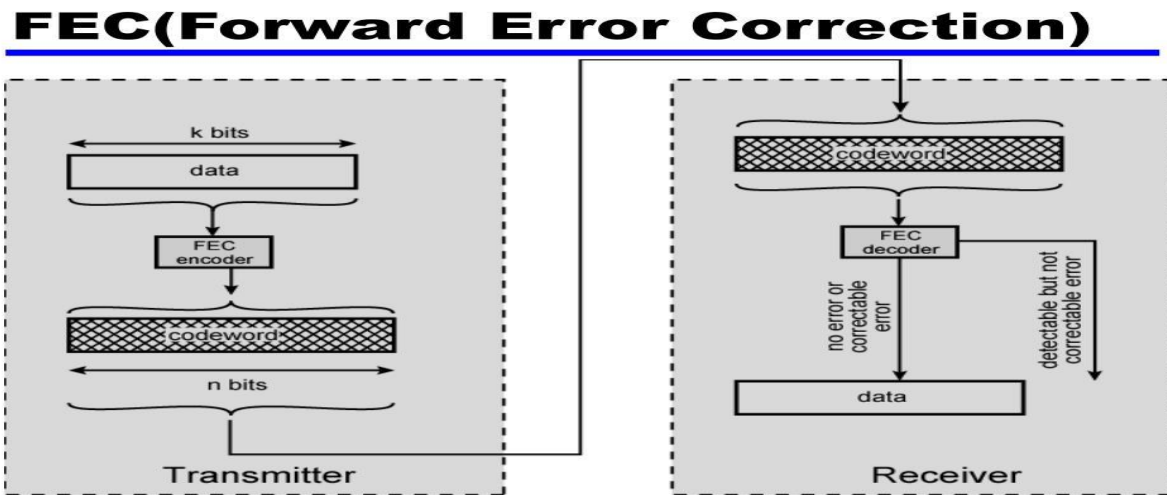


Figure 19:Forward Error Correction Process in PCIe [24]

2.7 Previous PCIe Implementations

Previous PCIe implementations, particularly those targeting **PCIe Gen3, Gen4, and PCIe Gen5**, have been widely explored in both academic research and industry documentation. Numerous open-source and commercial projects provide RTL models, verification environments, architectural analyses, and signal-integrity studies for these generations. As a result, design information for PCIe Gen4 and Gen5 is relatively accessible compared to newer generations. Several FPGA-based implementations of PCIe endpoints and root complexes have been reported, with a primary focus on **NRZ-based signalling**, protocol handling, and data-movement efficiency.

Representative examples of prior work include FPGA-based PCIe DMA architectures, such as the implementation presented in [19], which demonstrates a high-throughput PCIe design optimized for efficient data transfer between an FPGA and a host system. This class of work primarily addresses protocol-level behaviour and system-level performance using earlier PCIe generations, without considering advanced physical-layer techniques such as PAM4 modulation, Forward Error Correction (FEC), or FLIT-based framing.

Despite the abundance of literature and implementations for PCIe Gen3 through Gen5, there is a clear lack of dedicated, fully verified digital implementations of PCIe Gen6. Existing publications typically discuss PCIe Gen6 only at a theoretical or architectural level, without presenting complete digital models, RTL implementations, or comprehensive verification results that incorporate the fundamental changes introduced in Gen6.

A broader review of previous PCIe implementations further reveals that most designs focused on logical modelling and protocol functionality rather than the physical-layer challenges of ultra-high-speed links. Even FPGA-based Gen4 and Gen5 implementations reported significant timing-closure and signal-integrity challenges at 16 GT/s and 32 GT/s, highlighting the increasing difficulty of implementing PCIe systems as data rates scale.

Specifications	Lanes				
	x1	x2	x4	x8	x16
2.5 GT/s (PCIe 1.x +)	500 MB/S	1 GB/S	2 GB/S	4 GB/S	8 GB/S
5.0 GT/s (PCIe 2.x +)	1 GB/S	2 GB/S	4 GB/S	8 GB/S	16 GB/S
8.0 GT/s (PCIe 3.x +)	2 GB/S	4 GB/S	8 GB/S	16 GB/S	32 GB/S
16.0 GT/s (PCIe 4.x +)	4 GB/S	8 GB/S	16 GB/S	32 GB/S	64 GB/S
32.0 GT/s (PCIe 5.x +)	8 GB/S	16 GB/S	32 GB/S	64 GB/S	128 GB/S
64.0 GT/s (PCIe 6.x +)	16 GB/S	32 GB/S	64 GB/S	128 GB/S	256 GB/S
128.0 GT/s (PCIe 7.x +)	32 GB/S	64 GB/S	128 GB/S	256 GB/S	512 GB/S

Table 3:PCIe Lanes Implementation [13]

In summary, while earlier PCIe generations have been extensively implemented and studied, there is no publicly available academic or open-source work provides a complete digital PCIe Gen6 model integrating PAM4 signalling, FEC, and FLIT mode. This clear research gap motivates the focus of this project, which aims to propose and implement an educational yet comprehensive PAM4-based digital PCIe Gen6.0 model, addressing an area not covered by prior implementations.

2.8 Verification Challenges in PCIe Gen6

Verification complexity in PCIe Gen6 has increased significantly due to the combined impact of several major architectural changes, including PAM4 signaling, FLIT-based packet framing, lightweight Forward Error Correction (FEC), and updated Data Link Layer (DLL) mechanisms. Recent literature, such as “**PCIe Verification at High Speeds: Challenges, Solutions, and Future Trends**”, emphasizes that verifying modern PCIe interfaces now requires environments capable of validating both digital protocol correctness and high-speed signaling behavior, creating a verification challenge that did not exist in earlier PCIe generations.[16]

One of the primary challenges highlighted is the verification of PAM4 signaling behavior. PAM4 introduces higher raw Bit Error Rates (BER), reduced vertical eye openings, and strong dependence on equalization techniques. Traditional digital-only verification environments are insufficient to accurately model these effects, making it difficult to reproduce real-world error scenarios. Consequently, verification methodologies must include mechanisms to inject realistic PAM4-related impairments, such as symbol errors, timing jitter, and voltage-level distortions, into the digital simulation flow.

The introduction of FLIT Mode further increases the verification complexity. Since FLITs are fixed-length structures that may encapsulate multiple TLPs, DLLPs, and control information, verification must ensure correct segmentation, packing, alignment, and reassembly under all traffic conditions. Even small errors in FLIT boundary handling or ordering logic can lead to subtle data corruption or protocol violations that are difficult to detect through conventional testing.

In addition, the interaction between CRC checking and lightweight FEC introduces new corner cases that must be carefully verified. For example, corrected bit errors must not invalidate CRC checking logic, and packets that exceed the correction capability of FEC must be correctly detected and retried. Validating these behaviors requires extensive stimulus coverage, sophisticated scoreboarding, and precise monitoring of retry and replay mechanisms.

Another challenge identified in existing studies is the lack of unified verification frameworks that integrate PAM4 modeling, FEC behavior, FLIT encapsulation, and sequencing logic into a single coherent environment. Most published research addresses only individual aspects of PCIe Gen6, leaving a gap in comprehensive verification approaches, particularly for educational and early-stage prototyping purposes.

Overall, the literature consistently concludes that PCIe Gen6 verification is substantially more demanding than previous generations, as it combines analog-level signaling effects introduced by PAM4 with increased protocol-level complexity from FLIT-based framing and FEC. These challenges strongly motivate the development of simplified, educational digital models that preserve the fundamental behavior of PCIe Gen6 while remaining verifiable within a purely digital design and simulation environment—an objective directly addressed by the model proposed in this project.

2.9 Identified Research Gaps

This section presents a detailed analysis of the identified research gaps in PCIe technology based on the conducted literature review. The identified gaps are categorized into key subtopics to provide a structured and clear understanding of the areas that still require further investigation.

Although existing research on PCIe Gen6 provides a strong theoretical foundation—covering topics such as the transition to PAM4 signaling, the introduction of FLIT-based framing, and the role of low-latency Forward Error Correction (FEC)—the literature still lacks several essential elements required for practical understanding and educational use. Most published work focuses primarily on architectural concepts, protocol modifications, electrical challenges, and expected performance improvements. However, none of the reviewed studies present a complete, step-by-step educational model that demonstrates how PCIe Gen6 operates across all protocol layers.

Specifically, there is no full system-level simulation that clearly illustrates the interaction between the Transaction Layer, Data Link Layer, and Physical Layer in a way that allows students or early-stage researchers to visualize packet flow, FLIT encapsulation, error detection and correction, replay mechanisms, and link behavior. In addition, despite the critical role of the PAM4-based Physical Layer in PCIe Gen6, there is no simplified digital Physical Layer model that abstracts complex analog behaviors into a form suitable for education, experimentation, or basic functional verification.

This gap significantly limits the ability of students and early-stage researchers to practically explore PCIe Gen6 behavior, validate theoretical assumptions, or evaluate architectural decisions through simulation. Addressing this limitation is a primary motivation for this project, which proposes a comprehensive educational modeling approach. The proposed model includes a simplified digital Physical Layer, a multi-layer system-level simulation, and clear demonstrations of key Gen6 mechanisms such as PAM4 signaling abstraction, FLIT processing, and FEC operation—combined into an accessible and practical learning tool.

In addition to the gaps described above, the literature also reveals the absence of a unified verification framework that simultaneously integrates PAM4 modulation, FLIT encapsulation, FEC decoding, and multi-layer sequencing behavior. Most existing studies examine these components in isolation, leaving a gap in understanding how they interact within a complete PCIe Gen6 system.

Furthermore, there is no academic-level performance estimation model for PCIe Gen6 that enables systematic analysis of throughput, latency, retry probability, and the impact of FEC. This project addresses these limitations by introducing a complete digital model capable of simulating the interaction between all three PCIe layers and their Gen6-specific mechanisms, thereby contributing both educational and practical value to the existing body of research.

Chapter Three: Proposed Methods

This chapter describes the design methodology for a simplified digital model of PCIe Gen6, structured into three layers: Transaction, Data Link, and Physical. The system block diagram illustrates the data flow from the Transaction Layer through the Data Link and Physical Layers, across the simulated channel between transmitter and receiver. Each layer's functionality is detailed, with the diagram highlighting how TLPs and DLLPs are processed, encapsulated into FLITs, and transmitted using PAM4 encoding. The chapter also presents the simulation plan, covering unit- and system-level testing, with emphasis on validating layer interactions and overall system performance.

3.1 System Architecture Overview

The proposed system represents a simplified digital model of a PCIe Gen6 communication link. The design follows the standard PCI Express layered architecture, which is composed of three main layers: Transaction Layer, Data Link Layer, and Physical Layer (Digital Model). The system is divided into two main sides: **the Transmitter side and the Receiver side**.

Between the two sides, a **simulated communication channel** is placed at the Physical Layer level to represent the physical connection between the transmitter and the receiver.

In this project, the Physical Layer is not implemented as a real analog hardware interface. Instead, it is modeled as a **behavioral digital layer** that simulates the main functions of a PCIe Gen6 physical interface, including FLIT formation, error protection, and PAM4 logical encoding.

The overall data flow in the system is as follows:

**Transaction Layer → Data Link Layer → Physical Layer (FLIT + FEC + PAM4) → Channel
→ Physical Layer → Data Link Layer → Transaction Layer**

This structure reflects the actual concept of PCIe Gen6, where Transaction Layer Packets (TLPs) and Data Link Layer Packets (DLLPs) are encapsulated into fixed-length FLITs at the Physical Layer before being transmitted over the physical channel.

To ensure a modular and scalable framework, the system architecture applies **layer isolation principles**, meaning each layer is designed, simulated, and verified independently before integration.

The architecture, as shown in **Figure20**, also incorporates **internal debug interfaces** that allow waveform inspection and event tracing during simulation. These debugging signals function similarly to the embedded logic analyzers used in commercial PCIe controllers, enabling step-by-step verification of TLP routing, sequence numbering, FEC operation, and PAM4 encoding.

3.2 Block Diagram of the Whole System

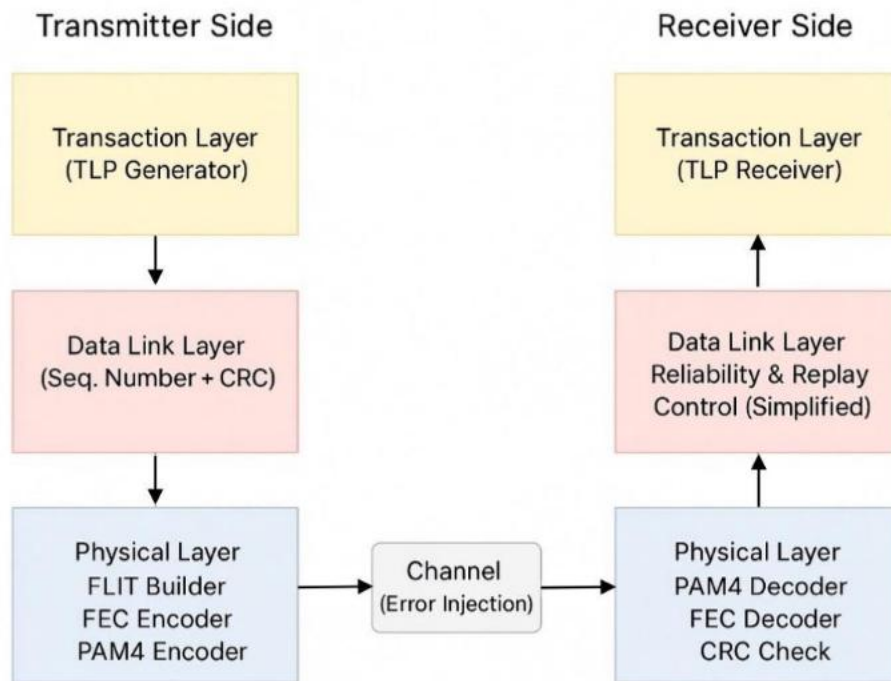


Figure 20:Simplified Block Diagram of PCIe Gen 6

The proposed PCIe Gen6 digital model follows a structured three-layer architecture that faithfully reflects the real PCI Express protocol while maintaining a simplified digital environment suitable for academic simulation. The system is divided into a transmitter side and a receiver side, separated by a behavioral digital channel. Each side contains the Transaction Layer, the Data Link Layer, and the Physical Layer, and the data flows sequentially through these layers in a complete end-to-end pipeline. The goal of this architecture is to model how PCIe Gen6 handles packet creation, reliability control, fixed-size FLIT framing, FEC protection, PAM4 logical encoding, lane bonding, lane alignment, and eventual TLP reconstruction at the receiver.

Data flow begins at the **Transmitter Transaction Layer**, where Transaction Layer Packets (TLPs) are constructed. The Transaction Layer generates packet headers, payloads, addressing information, and attributes that define the type and properties of each transaction. These TLPs are then passed downward to the Data Link Layer using a clean and structured interface. At this stage, the packets are still in their pure logical form, without framing or encoding, and represent the highest conceptual level of the PCIe communication path.

Upon entering the **Transmitter Data Link Layer**, each packet assigns a sequence number that uniquely identifies its position within the stream. This sequence number will later be used by the receiver to detect missing, duplicated, or out-of-order packets. The Data Link Layer also implements reliability mechanisms through the replay buffer, which stores outgoing TLPs until they are positively acknowledged by the receiver. Error detection mechanisms are applied by generating a Link CRC (LCRC) or a simplified CRC in accordance with the digital model. Once the packet is protected and numbered, it forms a complete DLL-formatted unit that is prepared for FLIT encapsulation in the Physical Layer.

The **Transmitter Physical Layer** is the core of the PCIe Gen6 behavior implemented in this project. Here, the packet is placed inside a fixed 256-byte FLIT, in FLIT mode, we are using a completely new TLP Header format. Previous TLP Headers had many limitations, like no room for increasing tag size. PCI-SIG redesigned header to suites suited FLIT mode. The challenge will be to test all the new combinations.

TLP Header is composed of a 3 to 7 DW TLP header Base, followed by 0 to 7 additional DWs of OHC (Orthogonal Header Content) as shown in **Figure21**.

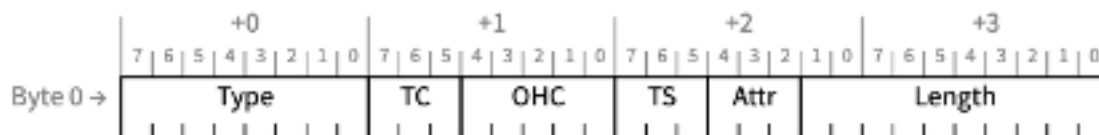


Figure 21: First DW of Header Base [20]

The new type of field has a fully decoded 8b Packet Type field. This means that all the 256 Type values are defined or reserved for a certain group to permit proper framing and forwarding. Also, there are new completion rules designed for FLIT mode. The completion for non-posed TLPs also has major updates including a 14-bit tag, error report using OHC-A5, etc. as shown in **Figure22**.

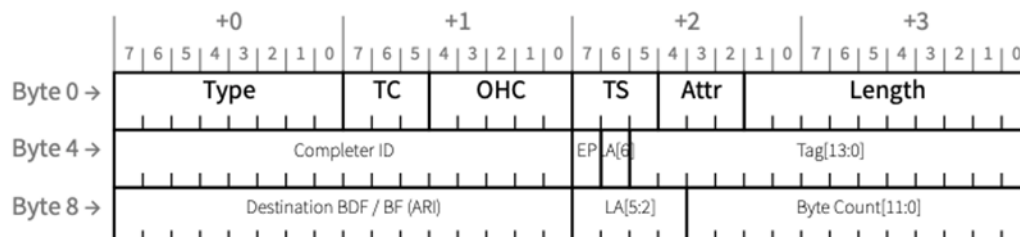


Figure 22:Completion Header Base Format - Flit mode [20]

the mandatory transmission format introduced in PCIe Gen6. The **FLIT Builder** collects TLPs and DLLPs, segments them if necessary, and assembles them into properly formatted FLITs that contain headers, metadata, control fields, and optional padding. Following **FLIT** construction, a lightweight **Forward Error Correction (FEC)** encoder is applied. This encoder generates parity bits that allow the receiver to correct symbol-level errors caused by the channel, which is especially important in Gen6 due to the transition to **PAM4** signaling. After that, the binary stream undergoes digital **PAM4** mapping, where every two bits are encoded into a single four-level symbol using gray coding, imitating the multi-level modulation used in real PCIe Gen6 links.

A crucial part of the Physical Layer is the **Lane Bonding mechanism**, which distributes the PAM4 symbols across the available lanes (x1, x4, x8, x16). The transmitter stripes the FLIT data across multiple lanes in parallel, ensuring maximum throughput and correct alignment with the link width. Lane Bonding includes several digital operations such as lane ordering, lane indexing, and the generation of alignment markers so that the receiver can reassemble the data correctly even if lanes experience different delays. Each lane carries its corresponding symbol stream toward the digital channel model, completing the behavior of the Transmitter Physical Layer.

The **digital channel** represents the path between transmitter and receiver but is modeled entirely using digital logic. It introduces controlled impairments such as bit errors, symbol flips, random delay, and lane-to-lane skew. These impairments allow the FEC, CRC, disked logic, and replay mechanism to be tested in realistic conditions without simulating full analog signal integrity. The channel ensures that the system is stress-tested at the level expected of a PCIe Gen6 behavioral model.

On the receiving side, data first enters the **Receiver Physical Layer**, where the reverse operations occur. The incoming lanes are realigned using lane deskew buffers, which ensure that all lane data is synchronized before reassembling FLITs. Lane Bonding is applied in reverse, allowing the receiver to identify lane indices, correct lane ordering, and handle polarity inversion if necessary. After lane alignment, PAM4 logical decoding converts each symbol back into two binary bits, reconstructing the original serialized stream. FEC decoding is then applied to correct symbol-level errors injected by the channel. Once the error-corrected stream is available, the FLIT Parser extracts TLPs, DLLPs, metadata, and control information from the 256-byte FLIT structure and passes them upward to the Data Link Layer.

The **Receiver Data Link Layer** then verifies the sequence number to maintain correct packet ordering and detect missing transmissions. It checks the CRC or LCRC to confirm packet integrity. If a packet fails to verification or contains uncorrectable FEC errors, a NAK is generated to trigger replay from the transmitter's replay buffer. If the packet is valid, an ACK is issued, and the replay buffer entry at the transmitter is safely released. This ensures full reliability of the communication channel, maintaining the fundamental operational behavior of PCIe links.

Finally, the packets reach the **Receiver Transaction Layer**, where the TLPs are fully reconstructed. The Transaction Layer parses the header, extracts the payload, and delivers the completed request or completion to the upper system logic. This marks the end of the complete **TX-to-RX pipeline**, demonstrating how PCIe Gen6 packets are generated, transmitted, protected, encoded, striped, deskewed, corrected, verified, and finally delivered—all inside a coherent digital simulation environment.

3.3 Decision Feedback Equalization (DFE)

Decision Feedback Equalization (DFE) as shown in **Figure 23** is a technique used to reduce the impact of intersymbol interference (ISI) on high-speed communication channels. **ISI** occurs when previously transmitted symbols leave residual effects that distort the current symbol, making it harder for the receiver to correctly interpret the incoming data.

The **DFE** method addresses this issue by using the decisions made on previously detected symbols. These past decisions are fed back into the equalizer to estimate the amount of ISI expected from earlier symbols. The estimated interference is then subtracted from the incoming signal, effectively “cleaning” the waveform before deciding the present symbol. By removing this residual distortion, DFE significantly improves the accuracy of symbol detection.

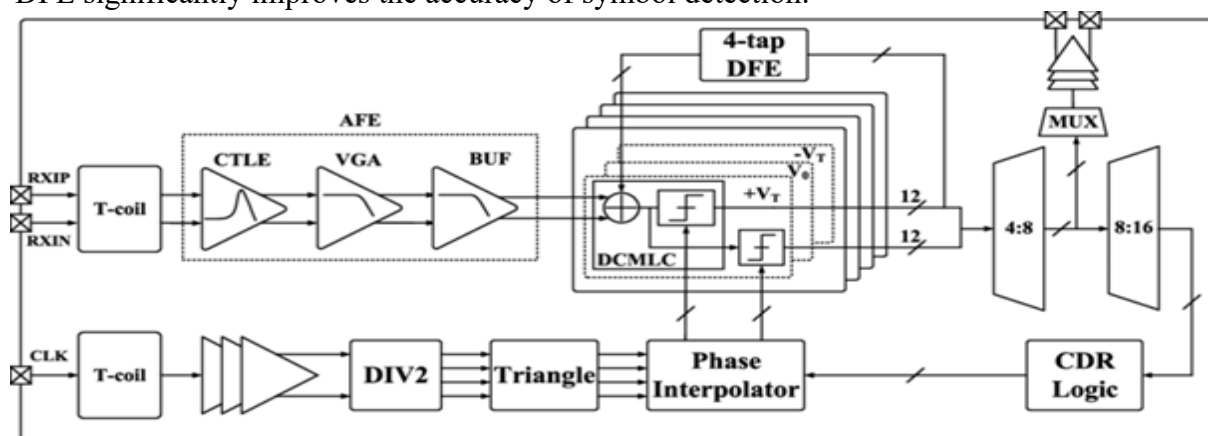


Figure 23: Overall DFE Block Diagram (Core Architecture) [21]

Validating **DFE** for PCIe 6.0 requires verifying that the equalization technique performs correctly at extremely high data rates. The validation process ensures that DFE can reliably compensate for ISI under the fast and complex signaling conditions of the PCIe 6.0 channel. It also confirms that the equalization does not introduce additional noise, distortion, or instability that could degrade link performance.

DFE behavior depends heavily on previous symbol decisions and channel conditions; accurate modeling is essential. The validation must capture the dynamic nature of the feedback process to ensure that DFE consistently mitigates ISI without generating unwanted side effects. This detailed modeling helps guarantee that the DFE operates effectively and supports the stringent performance requirements of PCIe 6.0.

3.3.1 Input Signal from the Channel (ISI)

At very high speeds (PCIe Gen 6 PAM4), symbols overlap in time, this overlap is called Inter-Symbol Interference (ISI). The current symbol is distorted by previous symbols. Without equalization, correct symbol detection becomes difficult. **Figure24**

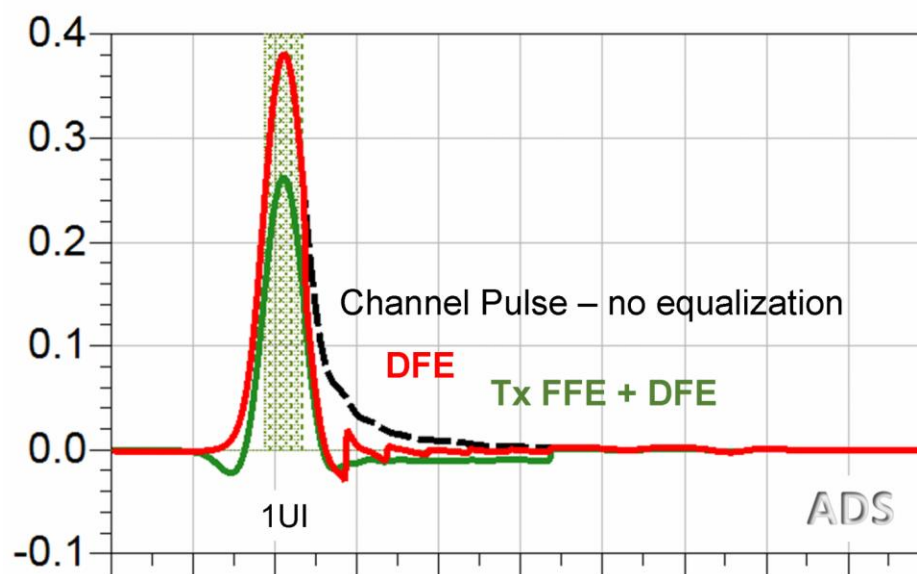


Figure 24:ADS IBIS simulation of a pulse response with and without equalization [22].

3.3.2 Feed-Forward Equalizer (FFE)

The **FFE is a linear filter**, it mitigates **pre-cursor ISI** (interference coming *before* the current symbol), It improves the signal shape **before slicing**, FFE alone cannot remove **post-cursor ISI**, which is why DFE is needed

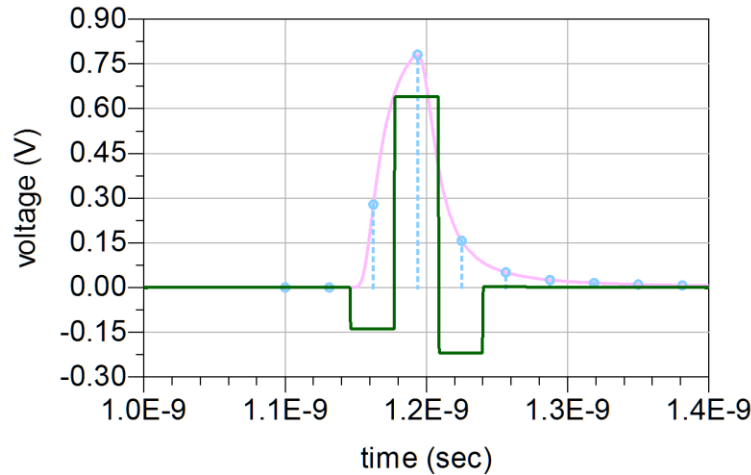


Figure 25:FFE inserts pulses of negative amplitudes to cancel out ISI that has positive amplitude.[21]

3.3.3 Decision Device (Slicer)

PAM4 receiver with adaptive threshold voltage, adaptive decision feedback equalizer, and fixed linear equalizer have been presented. The proposed techniques enable threshold voltage to be adjusted automatically depending on data rate, signal swing, and loss of channel. Consequently, the receiver can be used in various situations without being manually calibrated. Adaptive decision feedback equalizer for PAM4 signaling is proposed, incorporated with a sign-sign least mean square algorithm. Simulation results across a lossy channel show proper convergence of threshold voltage and decision feedback equalizer values with the proposed receiver.

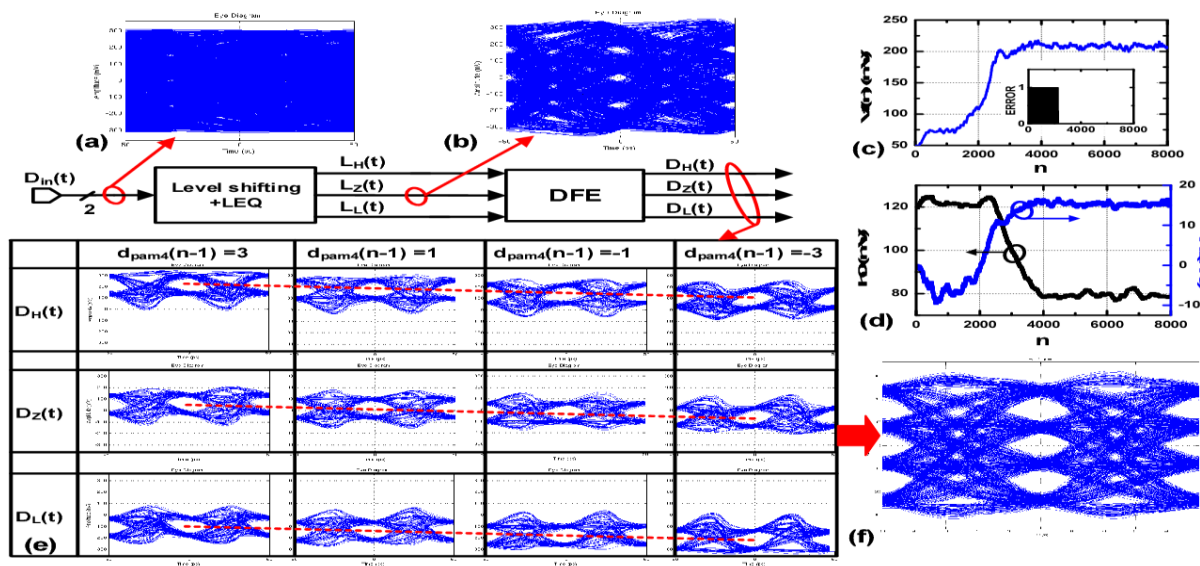


Figure 26:Simulation results of the proposed PAM4 receiver. [23]

Eye diagram of (a) D_{in} (b) L_L ; (c) Convergence of V_t ; (d) Convergence of h_0 and h_1 ; (e) eye diagrams of D_H , D_Z , and D_L ; (f) eye diagram of the reconstructed signal after DFE.

3.4 design and Simulation Plan

The design and simulation plan of the proposed PCIe Gen 6 digital model follows a structured, simulation-driven approach. The system is implemented using Verilog HDL, enabling modular and deterministic modeling of protocol components such as FLIT framing, CRC, FEC, and logical PAM4 processing. Verification is carried out using ModelSim, where both unit-level and system-level simulations are performed. Individual blocks are first tested independently using directed test vectors to ensure correct functionality, after which the complete transmitter-to-receiver pipeline is simulated to validate end-to-end behavior. Controlled error injection is used within the digital channel model to evaluate reliability mechanisms and observe system response under fault conditions. Simulation waveforms and internal signals are analyzed to extract performance metrics such as latency, throughput efficiency, and error-correction effectiveness. This approach provides a reliable and efficient method for validating the design and obtaining meaningful simulation results in an academic environment **Figure27**.

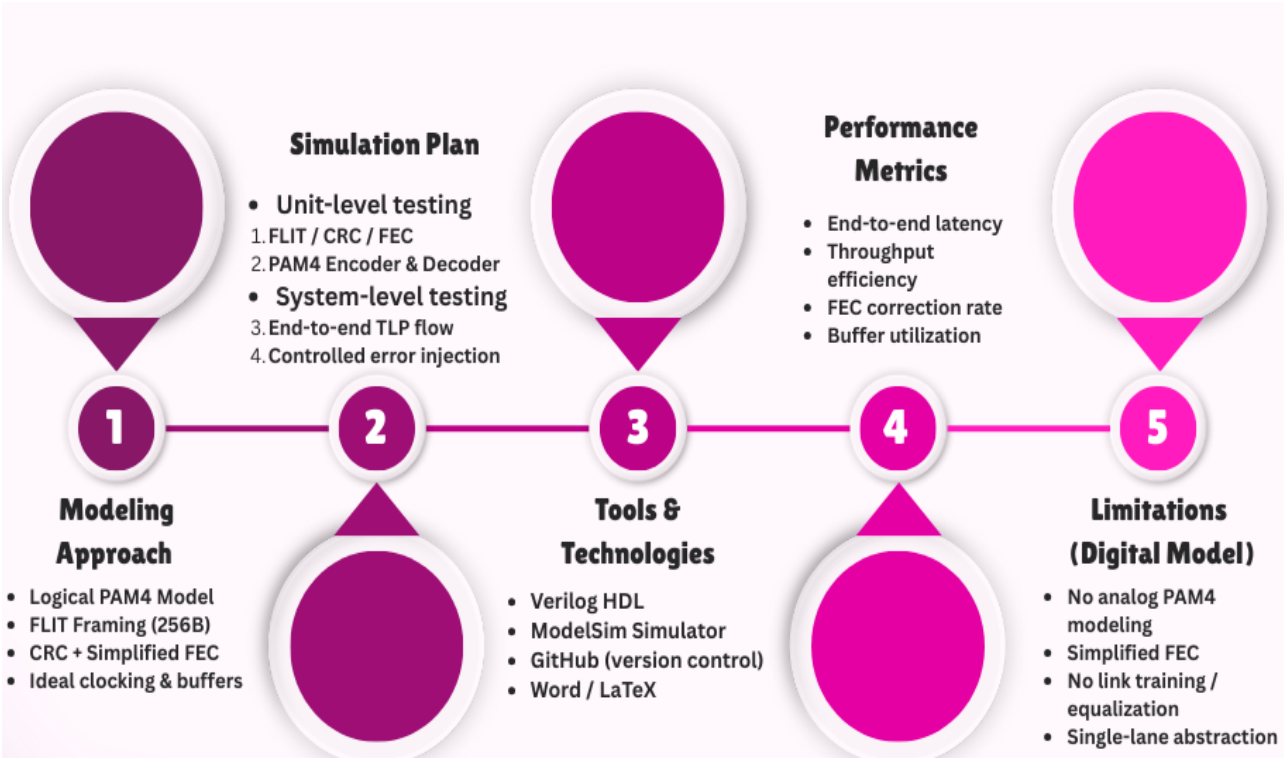


Figure 27:Design and Simulation Plan for the Proposed PCIe Gen6 Model

3.4.1 Modeling Approach

The proposed PCIe Gen6 digital model is developed under ensuring clarity, modularity, and feasibility within an academic environment. One of the primary roles is the abstraction of all analog behaviors typically associated with PAM4 signaling. Instead of simulating real analog waveforms, the design implements a logical PAM4 representation that focuses on symbol mapping, encoding behavior, and multi-level data transitions. This abstraction allows the model to preserve conceptual correctness without requiring complex analog simulation tools.

Another design assumption involves the use of simplified lightweight FEC architecture. While real PCIe Gen6 employs an advanced low-latency FEC optimized for high-speed links, the model uses a reduced version that maintains the core principles of error detection and correction. These constraints ensure that the design remains computationally manageable while still demonstrating the interaction between FEC, CRC, and FLIT framing.

Physical channel impairments such as jitter, inter-symbol interference (ISI), crosstalk, and channel loss are not included explicitly. Instead, controlled error injections are used to approximate their effect in a digital environment. This approach provides a balance between realism and educational simplicity. Additionally, the system assumes ideal clocking, synchronous operation, and immediate buffer availability — constraints that significantly reduce architectural overhead and keep the design focused on protocol behaviors rather than hardware implementation challenges.

3.4.2 Simulation Plan

within a comprehensive testbench environment. The verification process begins at the unit level, where individual functional blocks are tested independently to ensure correct operation before integration into the full system. These blocks include the FLIT Builder and Parser, responsible for constructing and deconstructing 256-byte FLITs; the CRC generator and checker, which detect data corruption; the FEC encoder and decoder, providing symbol-level error correction for PAM4 signals; the PAM4 encoder and decoder, mapping binary data to four-level symbols and back; and the Data Link Layer reliability logic, which manages sequence numbering, replay buffers, and acknowledgment handling. Each block is exercised with targeted test vectors to validate its behavior in isolation.

After unit-level testing, the system-level verification evaluates the complete end-to-end pipeline from transmitter to receiver. Transaction Layer Packets (TLPs) are transmitted through the behavioral digital channel, and correct data recovery is observed at the receiver. Controlled errors are injected into the channel to test the robustness of the FEC and CRC mechanisms and to verify proper operation of the replay and flow control logic. The simulation environment also measures end-to-end latency from TLP creation to reception, ensuring that buffer operations and lane alignment do not introduce excessive delays. High-throughput scenarios are simulated to evaluate buffer stability, sequencing logic, and overall system performance under stress conditions.

The impact of FLIT framing, CRC, and FEC overhead on effective throughput is analyzed to compare the simulated data rate with theoretical PCIe Gen6 expectations. This methodology provides a comprehensive verification framework that guarantees the PCIe Gen6 system maintains data integrity, responds correctly to errors, and meets performance targets, establishing a solid foundation for subsequent RTL implementation and further analysis.

3.4.3 Tools and Technologies

The development process of the PCIe Gen6 digital model relies on a set of tools and technologies commonly used in modern hardware design and verification of workflows. **Verilog HDL** is employed as the primary modeling language due to its structured syntax, hardware-orientated semantics, and suitability for modular digital design, **Figure28**. Its deterministic behavior allows precise modeling of protocol operations such as FLIT encapsulation, sequencing, and error handling.

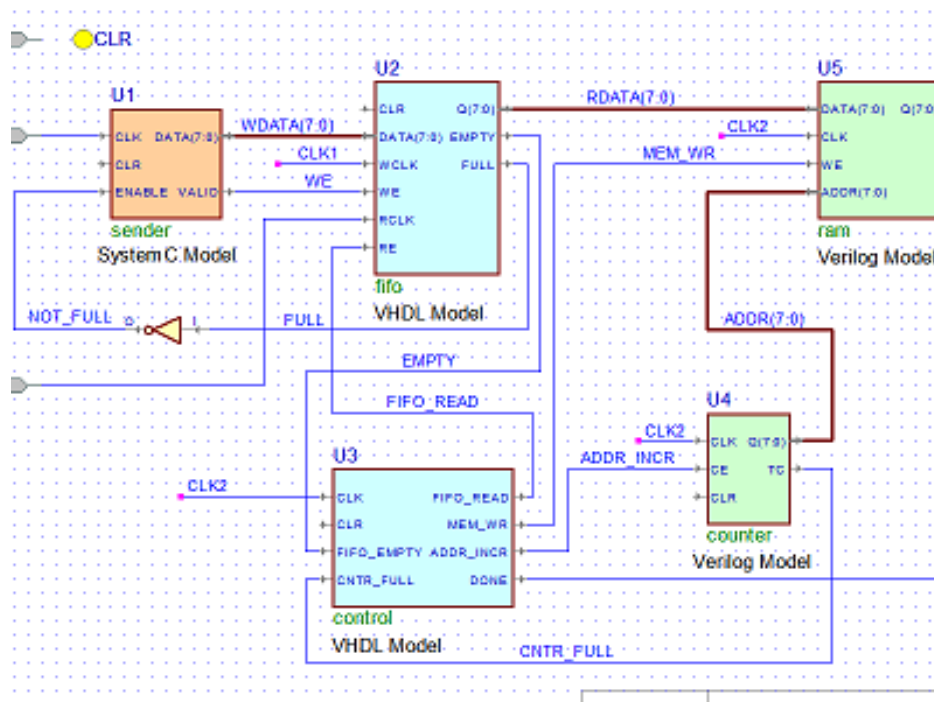


Figure 28:Schematic / Block Diagram Editor

Simulation and verification are conducted using **ModelSim**, an industry-standard tool that provides waveform visualization, timing analysis, and debugging capabilities. **ModelSim** allows step-by-step inspection of internal signals, replay-buffer behavior, and data flow across layers, making it ideal for protocol-level debugging.

Version control and collaboration are managed using GitHub, which ensures organized code development, tracking of design iterations, and effective teamwork. Documentation tools such as Microsoft Word and LaTeX were used to prepare project reports, ensuring professional formatting of diagrams, tables, and technical explanations, for example, Use Host Bridge to control debug a SoC using useful scripts (**Figure29**).

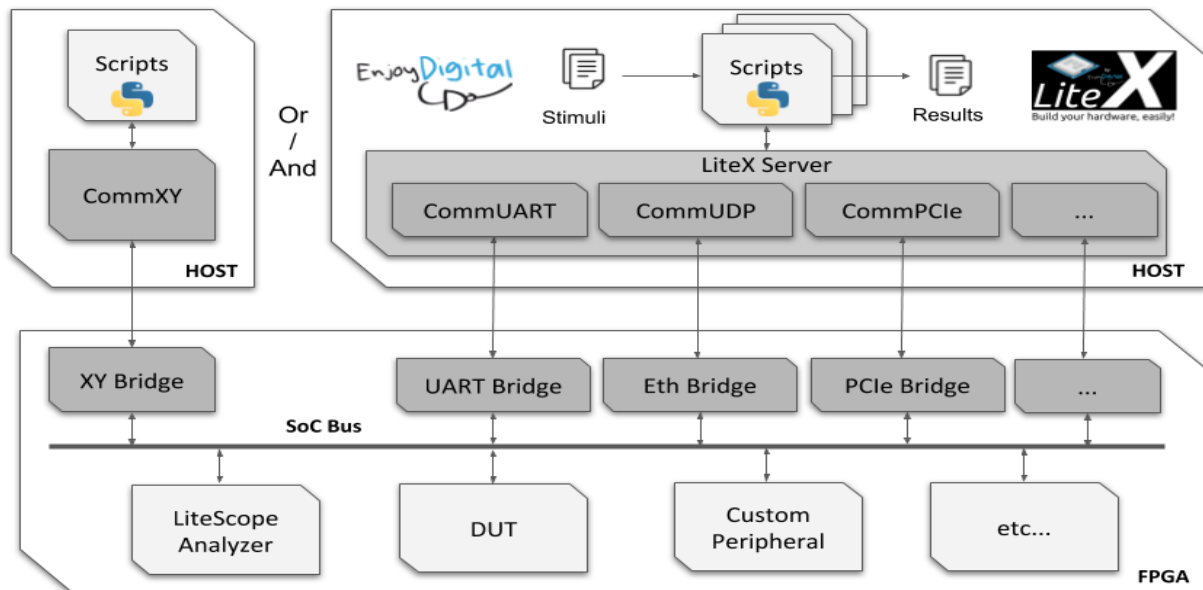


Figure 29:Example from GitHub

Although analog simulators such as SPICE or Cadence tools would normally be required for PAM4 waveform modeling, the constraints of this project make pure-digital tools sufficient. The toolchain used strikes a balance between practicality and academic value, providing accessible yet powerful means for understanding the complexity of PCIe Gen6.

3.4.4 Performance Metrics

To evaluate the effectiveness, correctness, and behavior of the proposed digital PCIe Gen6 model, several performance metrics are defined. The first key metric is **end-to-end latency**, which measures the number of cycles required for a packet to travel from the Transaction Layer of the transmitter to the Transaction Layer of the receiver. This metric reflects the combined effects of FLIT assembly, serialization, link traversal, buffering, and decoding.

Another metric is **throughput efficiency**, calculated as the ratio of useful payload to total transmitted bits, including overhead introduced by FLIT framing, PAM4 mapping, CRC, and FEC. This measurement is essential for comparing the model against the theoretical bandwidth of PCIe Gen6 links.

Error-correction effectiveness is also evaluated by measuring the number of injected errors that are successfully corrected by the simplified FEC module. This metric provides insight into how PAM4-related impairments affect reliability. Additionally, **sequence number integrity** is monitored to ensure correct ordering, replay behavior, and loss of handling across the Data Link Layer.

Finally, **buffer utilization metrics** are recorded to assess how efficiently the design handles bursty traffic, congestion, and high-load scenarios. These metrics collectively provide a comprehensive view of system behavior and allow meaningful comparisons with real PCIe Gen6 performance expectations.

3.4.5 Limitations Digital Model

Although the proposed model captures the fundamental characteristics of PCIe Gen6, several limitations exist due to the simplified digital approach. The most significant limitation is the absence of true analog modeling. Real PAM4 behavior relies on multi-level voltage signals, channel loss models, equalization stages, and clock recovery circuits — none of which are fully represented in the digital abstraction used in this project. As such, the model cannot predict analog-level signal integrity or BER performance.

Another limitation is the simplified FEC structure. The real PCIe Gen6 specification uses a carefully engineered low-latency FEC optimized for short correction windows and minimal data overhead. The digital model uses a reduced version that demonstrates the concept but does not replicate the actual code structure, decoding complexity, or latency characteristics of real hardware.

The design also omits link training procedures, adaptive equalization, and lane margining mechanisms that are mandatory in high-speed PCIe systems. These mechanisms are extremely difficult to replicate in a pure digital environment without analog feedback loops. Furthermore, the model does not capture multi-lane behavior, lane bonding, polarity inversion, or advanced replay scenarios involving multi-FLIT pipelines. The absence of these features limits the design's fidelity but keeps it practical for educational use. Moreover, these limitations are intentional, as the goal of the project is to provide a clear, structured, and accessible digital representation of PCIe Gen6 not to produce a full industry-grade implementation.

Chapter Four:

CONCLUSION AND

FUTURE WORK

4.1 Conclusion

This work presents a structured and educationally oriented architectural model of PCI Express (PCIe) Gen6, addressing the growing need for simplified yet accurate representations of modern high-speed interconnect standards. The proposed design delivers complete end-to-end architecture spanning the Transaction Layer, Data Link Layer, and Physical Layer, while incorporating key Gen6 enhancements, including FLIT-based data transfer, PAM4 signaling, and forward error correction (FEC) with CRC protection. By abstracting these advanced features into a coherent and modular framework, the project bridges the gap between highly theoretical specifications and implementation-focused academic study.

A major contribution to this work lies in overcoming the limited availability of clear academic reference models for PCIe Gen6. Most existing literature emphasizes prior generations or presents Gen6 concepts in a fragmented and highly theoretical manner. To address this gap, the project's scope, objectives, and constraints were carefully defined at an early stage, enabling a focused research direction and ensuring alignment between theoretical analysis and architectural design.

The inherent complexity of PCIe as a layered protocol posed a significant research challenge, particularly in integrating Gen6-specific mechanisms such as FLIT mode, PAM4 modulation, and FEC within a unified system model. This challenge was addressed through an extensive review of PCIe architectural evolution and a detailed layer-by-layer analysis. Decomposing the system into well-defined functional blocks enabled a clear understanding of inter-layer interactions and facilitated the development of consistent and scalable architecture.

One of the key outcomes of this research is the development of a complete three-layer system architecture that clearly defines data flow from Transaction Layer Packet (TLP) generation through Data Link processing to Physical Layer transmission. A detailed system block diagram was constructed to represent packet sequencing, error detection and correction, FLIT formation, and PAM4 modulation. Clearly defined interfaces and responsibilities across layers enhance modularity, reduce design ambiguity, and support future extensions of the model.

In addition, a structured simulation and evaluation framework was established to assess system performance. Key performance metrics, including bit error rate (BER), latency, throughput, and link reliability, were identified, along with targeted test scenarios focusing on FLIT efficiency, FEC performance, and PAM4 behavior. This evaluation methodology provides a solid foundation for future RTL implementation, detailed simulation, and hardware validation, while minimizing integration risks.

Overall, this project contributes to a research-driven, implementation-ready educational model of PCIe Gen6 that supports both academic learning and practical exploration. The proposed architecture serves as a scalable platform for future research into advanced PCIe features, verification methodologies, and high-speed serial communication systems.

4.2 Future Work

The upcoming phase of this project will concentrate on the RTL realization of the proposed three-layer PCIe architecture. Independent modules will be implemented for the Transaction, Data Link, and Physical layers to ensure clear functional separation and structured system development.

Each layer will incorporate its core functionalities, including Transaction Layer Packet generation, error detection and correction mechanisms, FLIT-based data organization, and PAM4 signaling techniques suitable for high-speed operation.

A modular and parameterized design strategy will be adopted to support scalability, reusability, and simplified debugging, allowing the architecture to be easily extended or modified for future PCIe generations.

To enhance PCIe Gen6 performance and narrow the gap toward Gen7, latency-oriented optimizations will be integrated, such as streamlined FLIT processing, reduced buffering stages, and more efficient flow control and acknowledgment handling.

Verification will be addressed through the development of comprehensive test benches targeting both individual modules and full system integration, with test scenarios covering normal traffic conditions as well as injected fault and error cases.

Error management will be further improved by incorporating adaptive or lightweight forward error correction techniques that provide robust protection while minimizing latency and performance overhead at ultra-high data rates.

System performance will be evaluated using key metrics including throughput, latency, and bit error rate, with measured results compared against analytical and theoretical models to guide design optimization and efficiency improvements.

Finally, the architecture will be extended to support intelligent multi-lane operation, dynamic lane management, and application-specific configurations optimized for low-latency or high-throughput environments, in addition to FPGA-based prototyping and thorough documentation to support future academic and research-oriented development.

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